Chinese expert consensus on conversion and perioperative therapy of primary liver cancer (2024 edition)

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Background: A significant portion of primary liver cancer patients in China are diagnosed at intermediate-to-advanced stages, often making them ineligible for curative surgery. Furthermore, high postoperative recurrence rates, reaching up to 70%, pose a major challenge for long-term survival. The emergence of novel systemic treatments, such as immune checkpoint inhibitor combinations, and advancements in locoregional therapies have created new opportunities for conversion and perioperative strategies. This updated consensus aims to standardize the clinical application of these therapies based on the latest evidence, with the objective of improving patient prognosis.

Methods: A multidisciplinary committee of 97 experts was convened to revise previous guidelines. The process involved a comprehensive search of medical databases and conference proceedings, with evidence graded according to the Grading of Recommendations Assessment, Development, and Evaluation (GRADE) system. Consensus statements were finalized through a formal electronic voting process, requiring at least 80% agreement for approval, resulting in 18 updated statements.

Results: The consensus provides refined definitions for conversion and perioperative therapy. It recommends various strategies for oncological conversion, including systemic therapy with anti-angiogenic drugs plus immunotherapy, and locoregional approaches like precision transarterial chemoembolization (TACE) and hepatic artery infusion chemotherapy (HAIC). The document strongly affirms surgical resection as a crucial step for achieving long-term survival after successful conversion and offers guidance on surgical timing and adjuvant therapy. For resectable patients with high-risk features, neoadjuvant and adjuvant treatments are outlined to mitigate recurrence. The consensus also advocates for using dynamic enhanced magnetic resonance imaging (MRI) and the modified Response Evaluation Criteria in Solid Tumors (mRECIST) criteria for efficacy assessment and underscores the essential role of a multidisciplinary team in management.

Conclusions: This updated consensus offers standardized, evidence-based guidance for clinicians on implementing conversion and perioperative strategies to optimize patient-centered care and highlights the need for continued research to further refine these promising approaches.

Keywords: Liver neoplasms; conversion therapy; perioperative therapy; consensus; China

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Introduction

With the publication of the "Chinese expert consensus on conversion therapy for hepatocellular carcinoma (2021 edition)", "Chinese expert consensus on conversion therapy of immune checkpoint inhibitors combined antiangiogenic targeted drugs for advanced hepatocellular carcinoma (2021 edition)", "Chinese expert consensus on neoadjuvant and conversion therapy for hepatocellular carcinoma", and "Chinese expert consensus on postoperative adjuvant therapy for hepatocellular carcinoma (2023 edition)", the concepts of conversion and perioperative therapy have been further promoted and applied in the clinical practice of liver cancer treatment (1). More high-level evidence has emerged in the fields of conversion and perioperative therapy, particularly research results adapted to China's conditions. To further clarify concepts, standardize treatments, and improve the prognosis of patients with primary liver cancer, Society of Oncology, Chinese Medical Association in collaboration with Liver Cancer Committee of China, Chinese College of Surgeons , Chinese Anti-Cancer Association, and The Liver Tumor Branch of the China International Exchange and Promotive Association for Medical and Healthcare have convened multidisciplinary experts from across the country to revise and update the "Chinese expert consensus on conversion and perioperative therapy of primary liver cancer (2024 edition)" (hereinafter referred to as the consensus).

Highlight box

Key recommendations

- The updated consensus emphasizes the importance of multidisciplinary collaboration in the conversion and perioperative treatment of hepatocellular carcinoma (HCC) to enhance patient outcomes.

What was recommended and what is new?

- It is recommended to utilize a combination of systemic therapies, locoregional treatments, and surgical interventions tailored to individual patient profiles to improve long-term survival rates.
- The 2024 edition introduces refined definitions and strategies for conversion and perioperative therapy, incorporating recent advancements in systemic anti-tumor therapies and locoregional treatments.

What is the implication, and what should change now?

- Healthcare providers should adopt a more integrated approach to managing HCC, emphasizing timely surgical interventions after successful conversion therapy to maximize patient survival outcomes.

Methods

This consensus was drafted by searching public databases including PubMed, EMBASE, Cochrane Library, and the Chinese Biomedical Literature Database (CBM), as well as abstracts from recent international conferences primarily sourced from the American Society of Clinical Oncology (ASCO) and the European Society for Medical Oncology (ESMO). The search terms included "hepatocellular carcinoma", "conversion therapy", "neoadjuvant therapy", "adjuvant therapy", "perioperative therapy", " transarterial intervention therapy", "systemic therapy", " locoregional therapy", and "radiotherapy", combining both medical subject headings (MeSH) terms and free-text terms in English and Chinese. The consensus was revised based on content from the "Chinese expert consensus on conversion therapy in hepatocellular carcinoma (2021 edition)", hot topics, and recent clinical applications by domestic experts.

This consensus evaluates the evidence level and recommendations by referring to the "National Health Commission of the People's Republic of China—the Guidelines for Diagnosis and Treatment of Primary Liver Cancer (2024 edition)". The evaluation of evidence-based medicine follows the principles outlined in the Grading of Recommendations Assessment, Development, and Evaluation (GRADE) system, using the "Levels of Evidence in 2011, Oxford Centre for Evidence-Based Medicine" as an auxiliary tool for specific execution of evidence grading (Levels 1 to 5), as detailed in Table S1. In converting evidence into recommendations, the consensus primarily follows the GRADE system for recommendation grading, while also incorporating modifications based on the ASCO guidelines grading (Table S2). The recommendations in this consensus are classified into three levels: A (strong recommendation), B (moderate recommendation), and C (weak recommendation).

The consensus utilized an electronic vote to calculate the strength of consensus. Votes were categorized into five levels: A for complete agreement; B for agreement with minor modifications; C for agreement with major modifications; D for neutral; and E for disagreement. The strength of consensus was calculated as follows: Consensus strength (%) = (Number of A + B experts) / Total number of experts × 100%. Consideration of consensus strength should balance patient benefits (efficacy and safety), accessibility and cost-effectiveness of treatment options,

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patient preferences, efficient resource utilization, and the level of evidence. A consensus ≥80% is considered achieved and approved; if <80%, further opinions are solicited at the discussion session, and post-discussion to draft the updated statement based on the results, which is then voted on again by correspondence. Initially, 20 consensus statements were preliminarily reached through discussion. On November 11, 2023, 97 expert group members participated in the discussion meeting and voted on the consensus for each statement, ultimately updating and achieving 18 consensus statements.

Current status of conversion and perioperative therapy for hepatocellular carcinoma (HCC)

Primary liver cancer is currently the fourth most common malignant tumor and the second leading cause of cancer-related death in China, posing a significant threat to the health and life of the Chinese population (2). HCC accounts for 75% to 85% of primary liver cancers. Radical resection remains the most effective treatment for achieving long-term survival in HCC. For early-stage HCC {China Liver Cancer Staging (CNLC) stages Ia [equivalent to Barcelona Clinic Liver Cancer (BCLC) 0/A], Ib (equivalent to BCLC A), and IIa (equivalent to BCLC B)}, the preferred treatments include surgical resection, local ablation, and liver transplantation, with median survival times exceeding 5 years. However, the 5-year postoperative recurrence and metastasis rate for HCC patients can reach up to 70% (3). About 64% of Chinese patients with HCC are diagnosed at an intermediate-to-advanced stage [CNLC stages IIb (equivalent to BCLC B), IIIa and IIIb (both equivalent to BCLC C)] with poor prognosis (4). Although a subset can still benefit from surgical resection, the majority are no longer suitable for surgery as the first-line treatment and require conversion and perioperative therapies among other multidisciplinary approaches to potentially improve long-term survival rates.

In recent years, significant progress has been made in non-surgical treatments for HCC. Systemic anti-tumor therapies, particularly the combined use of targeted drugs and immune checkpoint inhibitors (ICIs), such as atezolizumab [an anti-programmed death-ligand 1 (PD-L1) antibody] combined with bevacizumab (IMbrave150 study), sintilimab combined with a bevacizumab biosimilar (IBI305) (ORIENT-32 study), and camrelizumab combined with apatinib (CARES-310 study), have been used in advanced or unresectable HCC. These combinations have achieved objective response rates (ORRs) of 20% to 30% [based on Response Evaluation Criteria in Solid Tumors (RECIST) 1.1], with median survival time extended to about 20 months (5). Additionally, locoregional therapies such as transarterial chemoembolization (TACE), hepatic artery infusion chemotherapy (HAIC), and radiotherapy have also seen improvements through technological and pharmaceutical advancements. In combination with other treatment modalities, better outcomes and prognosis have been achieved in terms of tumor shrinkage, and tumor thrombus control compared to past approaches. Based on these explorations, the consensus editorial committee, characterized by multidisciplinary participation, coexistence of multiple approaches, and diversified conclusions, has combined existing approaches and credible data to form a roadmap for conversion and perioperative therapy of HCC (*Figure 1*).

Reassessment of definitions related to HCC conversion and perioperative therapy

Concept of conversion therapy: According to the "the Guidelines for Diagnosis and Treatment of Primary Liver Cancer (2022 Edition)" issued by the National Health Commission of the People's Republic of China (2), the fundamental principles of HCC surgical resection include: (I) radicality: complete tumor removal with no residual tumor at the margins; (II) safety: preservation of a sufficient volume of functional liver tissue (with good blood supply and adequate blood and bile flow) to ensure postoperative liver function compensation, reducing surgical complications and mortality. Based on these principles, patients who can undergo surgery following these guidelines are deemed suitable for surgical resection; otherwise, they are considered unsuitable. Conversion therapy refers to the process where patients initially unsuitable for surgery become candidates for resection following intervention, mainly involving conversion of the functional future liver remnant (FLR) and oncological conversion. This treatment is primarily applicable to patients in CNLC stages Ib (equivalent to BCLC A) to IIIa (equivalent to BCLC C) who are initially unsuitable for surgery but have the potential to become resectable.

Concept of perioperative therapy: Treatment surrounding the entire process of HCC surgery, aimed at further reducing the recurrence and metastasis rates after HCC resection and improving long-term survival. Perioperative treatment mainly includes neoadjuvant therapy, intraoperative therapy, and

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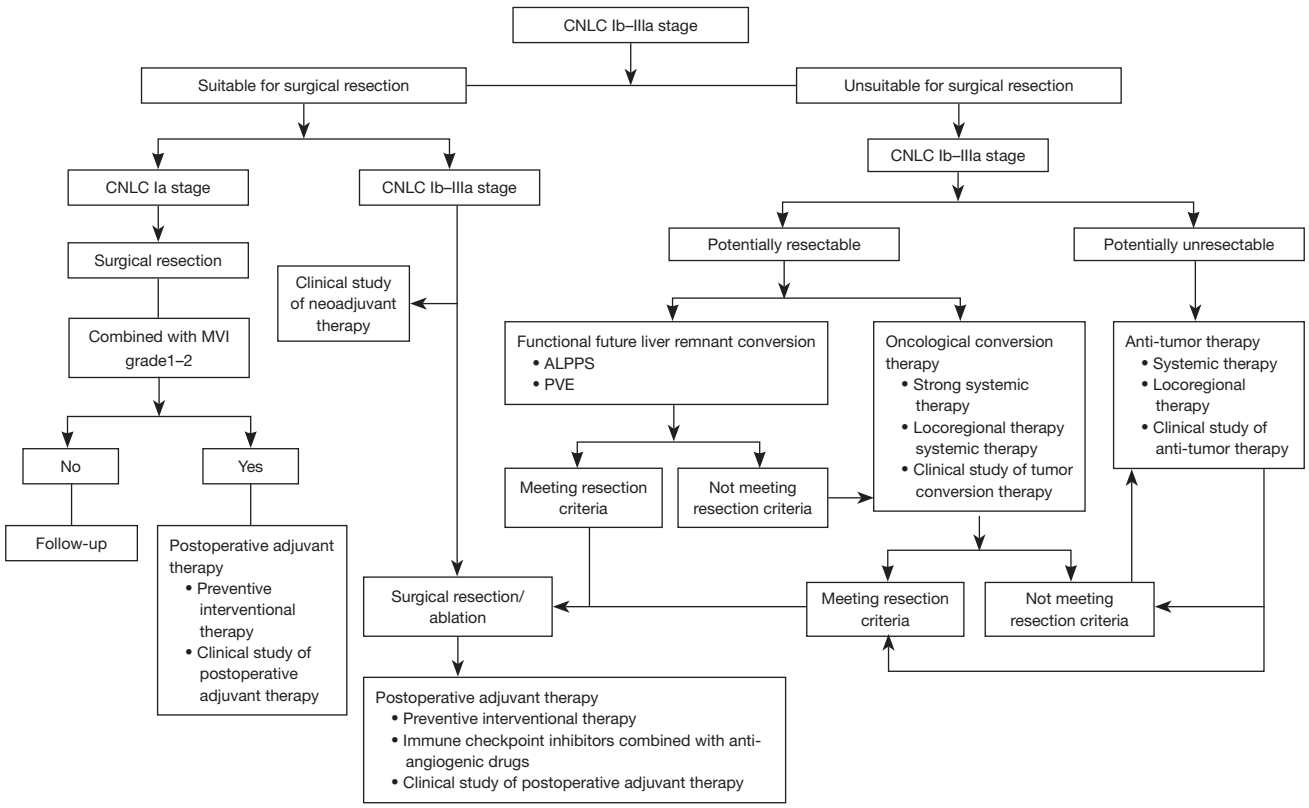


Figure 1 Treatment pathway of conversion and perioperative therapy of hepatocellular carcinoma. ALPPS, associating liver partition and portal vein ligation for staged hepatectomy; CNLC, China Liver Cancer Staging; MVI, microvascular invasion; PVE, portal vein embolization.

postoperative adjuvant therapy. For patients identified with high postoperative risks of recurrence and metastasis, such as those with microvascular invasion (MVI), macroscopic tumor thrombus, multiple tumors, satellite nodules, tumor diameter >5 cm, or adjacent organ involvement, aggressive perioperative therapy is recommended. Neoadjuvant therapy is mainly suitable for patients in CNLC stages Ib (equivalent to BCLC A) to IIIa (equivalent to BCLC C) who are candidates for surgery but have high risks of postoperative recurrence and metastasis. The applicability of postoperative adjuvant therapy is determined based on clinical characteristics, postoperative pathology, and molecular typing.

Both conversion therapy and perioperative therapy are intrinsically unified; their essence or ultimate goal is to achieve curative conversion for patients, thereby enabling long-term tumor-free survival.

Statement 1: conversion therapy refers to interventions aimed at making initially unresectable HCC patients eligible for surgical resection. These interventions primarily include functional FLR conversion and oncological conversion therapy.

Statement 2: perioperative therapy refers to the treatment processes surrounding the entire process of HCC surgery, aimed at further reducing the recurrence and metastasis rates post-resection and improving long-term survival rates. This includes neoadjuvant therapy, intraoperative treatments, and postoperative adjuvant therapy.

Functional FLR conversion in HCC conversion therapy

The prerequisites for surgical resection include liver function Child-Pugh Class A, indocyanine green retention rate at 15 min (ICG R15) <30%; the FLR should be larger than 40% of the standard liver volume (SLV) in patients with chronic liver disease, hepatic parenchymal damage, or cirrhosis, or more than 30% in patients without liver fibrosis or cirrhosis. In patients with impaired liver function,

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a greater proportion of FLR must be preserved. Insufficient FLR function is an important reason for the infeasibility of surgical resection in liver surgery. For such patients, the goal of conversion therapy is to convert insufficient FLR function into sufficient FLR function (2,6).

Portal vein embolization (PVE)

PVE involves embolizing the portal vein of the hemiliver containing the tumor to induce compensatory hypertrophy of the remaining liver before tumor resection. The conversion success rate of PVE is 60–80%, with a complication rate of 10–20%. The hypertrophy of the remaining liver after PVE typically takes 4–6 weeks, and about >20% of patients may lose the opportunity for surgery due to tumor progression or insufficient FLR hypertrophy during the waiting period (7,8). Current treatment strategies for these patients include combining TACE, hepatic vein embolization, and hepatic arterial ligation to further promote FLR hypertrophy and control tumor progression (9) or performing a salvage two-step hepatectomy, associating liver partition and portal vein ligation for staged hepatectomy (ALPPS), to remove the tumor (10). Contraindications for PVE include tumor thrombus in the main trunks or first-order branches of the portal vein, widespread tumor metastasis, severe portal hypertension, and coagulation disorders. PVE should be used cautiously in patients anticipated to have a long FLR hypertrophy duration (such as those with severe cirrhosis or older age), where tumor progression might occur more rapidly.

ALPPS

ALPPS is one of the significant innovations in the field of hepato-biliary surgery in recent years, usually inducing a 47–192% hypertrophy rate of the remaining liver within 1–2 weeks, which is higher than that of PVE, providing patients with insufficient FLR function short-term surgical opportunities. Due to the short interval between the two stages of surgery, it minimizes the risk of tumor progression, achieving tumor resection rates of 95–100% (11,12). Despite reductions in surgery-related complications and mortality rates with advances in surgical techniques and experience accumulation, attention is still warranted in managing complications associated with ALPPS. Recent years have seen various modifications to ALPPS, focusing primarily on the liver partition during the first stage [partial partition, and partition methods such as radiofrequency ablation (RFA), microwave, and tourniquet] and adopting laparoscopic minimally invasive approaches to further enhance the safety of ALPPS. ALPPS combined with salvage hepatic artery embolization can achieve nearly a 100% surgical resection rate in patients with mild-to-moderate cirrhosis (13). Recent randomized controlled trial (RCT) results show that ALPPS has a significant advantage over PVE in inducing FLR hypertrophy (14). ALPPS should generally be limited to patients younger than 65 years, with normal liver function (Child-Pugh Class A, ICG R15 <20%), insufficient FLR volume (FLR/SLV <30% in normal livers; <40% in those with chronic liver disease and liver injury), good general condition, good surgical tolerance, without severe cirrhosis, severe steatohepatitis, or severe portal hypertension.

Statement 3: functional FLR conversion refers to the short-term increase of the FLR function in HCC patients with insufficient FLR volume. Specific methods include ALPPS and PVE, with ALPPS having a higher conversion efficiency (Level of evidence 2, Recommendation A).

Oncological conversion in HCC conversion therapy

Application of systemic antitumor therapy in oncological conversion

Systemic antitumor therapy for HCC has evolved through chemotherapy, targeted therapy, ICIs, and combinations of targeted therapy and immunotherapy. Currently approved in China for first-line systemic antitumor therapy in advanced HCC, targeted drugs include sorafenib, lenvatinib, and donafenib; the approved chemotherapy regimen is the FOLFOX4 involving oxaliplatin; approved ICIs include atezolizumab, camrelizumab, sintilimab, and tislelizumab. Given the superior short and long-term efficacy of ICI-based combination therapies over monotherapy with ICIs, the guidelines and consensuses currently recommend PD-1 and its ligand (PD-L1) monoclonal antibodies combined with anti-angiogenic agents and PD-L1 monoclonal antibodies combined with cytotoxic T lymphocyte-associated antigen-4 (CTLA-4) monoclonal antibodies as the preferred treatment options (15-17). These combination regimens include atezolizumab with bevacizumab, sintilimab with a bevacizumab biosimilar, camrelizumab with apatinib, and tremelimumab with durvalumab. Additionally, ongoing phase III clinical trials include ICIs combined

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with anti-angiogenic agents (penpulimab with anlotinib, toripalimab with lenvatinib or bevacizumab, HLX10 with HLX04, SCT-I10A with SCT510), ICIs combined with chemotherapy (camrelizumab with FOLFOX4), and ICIs combined with other ICIs (nivolumab with ipilimumab, sintilimab with IBI310), with some trials having completed enrollment and results are eagerly anticipated.

The enhancement of therapeutic efficacy in systemic antitumor therapy for intermediate-to-advanced HCC has further promoted the development of combined treatment modalities for early-to-intermediate HCC. Systemic antitumor therapy has become an important tool for conversion and perioperative therapy. An overview of systemic antitumor therapy regimens related to conversion therapy, based on public database searches, is presented in *Table 1*.

From the perspective of conversion therapy, the most critical factors in evaluating systemic antitumor therapy regimens are ORR and response patterns, including tumor progression rate, time to response, duration of tumor response, and the depth of response. Researchers both domestically and internationally vary in their choice of conversion therapy regimens. Due to drug availability and strong antitumor activity, scholars in China mostly adopt PD-1 monoclonal antibodies combined with tyrosine kinase inhibitors (TKIs) as the systemic antitumor therapy (*Table 1*). In international clinical practice, some non-virus-related HCC cases achieve conversion opportunities with lenvatinib alone; some advanced cases use atezolizumab combined with bevacizumab. Currently, due to varying patient populations and different regimens, the range of conversion success rates fluctuates, and choosing conversion therapy based on staging system remains underdeveloped. Based on the small sample reports in *Table 1*, from the perspective of choosing systemic antitumor therapy regimens, it is necessary to identify reasons for inoperability, focus on etiological management, pay attention to the duration and degree of tumor remission within a multi-disciplinary team (MDT) framework, closely monitor the toxicity of systemic antitumor therapy and its potential impact on post-conversion surgical resection, actively explore changes in the tumor immune microenvironment before and after conversion therapy, and vigorously conduct large-scale clinical studies with high-level evidence-based medicine to maximize patient benefits.

Statement 4: anti-angiogenic drugs combined with ICIs are an important approach for conversion therapy for HCC (Level of evidence 3, Recommendation B). It is recommended to focus on the duration and degree of tumor remission, closely monitor by MDT the toxicity of systemic antitumor therapy, and evaluate its potential impact on surgical resection after conversion (Level of evidence 3, Recommendation B).

Application of transarterial interventional therapy in oncological conversion

TACE

TACE is a commonly used treatment method for unresectable intermediate-to-advanced HCC and an effective traditional therapy for oncological conversion. Currently, a wide range of chemotherapeutic drugs are available for use during TACE, and it is recommended to select the drug and dosage based on the patient's tumor burden, body surface area, prior medications, and comorbidities (35). In addition to conventional TACE, the advent of drug-eluting beads-TACE (DEB-TACE) allows for the slow and sustained release of chemotherapy drugs while embolizing the vessels, playing an important role in the treatment of intermediate-to-advanced HCC (36). Patients who undergo surgery after TACE conversion have a 5-year survival rate of 25% to 57%, roughly comparable to that of initially resectable patients (37). A retrospective study involving 831 Chinese patients with HCC found that among 82 patients who achieved partial remission (PR) after TACE, those in the surgical resection group had a longer overall survival (OS) (49 months compared to 31 months, P=0.027) and a higher 5-year survival rate (26% compared to 10%, P=0.019) than those in the conservative treatment group (38). The ORR after TACE, particularly the PR rate greater than 90%, is a key factor affecting long-term survival following sequential surgical treatment (39). Multiple TACE procedures may cause liver function damage and decrease the response rate. Studies have shown that the efficacy of the first TACE is an important factor affecting long-term prognosis (40). If the disease progresses after ≥3 consecutive TACE treatments, it indicates TACE resistance, necessitating a prompt adjustment of the treatment plan. Precision TACE can reduce the heterogeneity of TACE to improve ORR (41). Combining TACE with other locoregional therapies can further enhance the conversion success rate. A study involving initially unresectable HCC patients treated with TACE combined with HAIC showed a higher conversion rate for TACE combined with HAIC than

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Table 1 Reports on conversion therapy related to systemic therapy in advanced HCC

Systemic antitumor therapy regimen	Publication year	Sample size (n)	CNLC/BCLC staging of successful conversion cases	Treatment duration (months [range])	Objective response rate	Conversion rate[§]
TKIs monotherapy						
Lenvatinib (18)	2023	5	3 cases with stage B/2 cases with stage C	3 [2–8] months	3/5[†]	MPR
Lenvatinib (19)	2022	9	2 cases with stage B/6 cases with stage C	8 [4–50] weeks	2/9[†]	88%
TKIs + immune checkpoint inhibitors						
Sintilimab + lenvatinib (20)	2023	36	8 cases with stage B/4 cases with stage C	4.4 months	66.7%[†], 36.1%[‡]	33%
Tislelizumab + lenvatinib (21)	2023	24	–	4 cycles	75.0%[†], 54.2%[‡]	70.8% (3 pCR cases)
PD-1 monoclonal antibody + lenvatinib (22)	2022	107	1 Ia case/3 Ib cases/6 IIb cases/17 IIIa cases/3 IIIb cases	90.5 days	50%[‡]	28% (10 pCR cases)
PD-1 monoclonal antibody + lenvatinib (23)	2023	56	2 cases with stage B/29 cases with stage C	–	53.6%[†], 44.6%[‡]	55.4%
TKIs + PD-1 monoclonal antibody (24)	2023	101	3 Ib cases/1 IIa case/ 5 IIb cases/11 IIa cases/ 4 IIIb cases	3.9 months	49.5%[†], 32.7%[‡]	23.8%
TKIs + PD-1 monoclonal antibody (25)	2022	187	9 IIb cases/10 IIIa cases/ 10 IIIb cases	–	37.4%[†]	15.5%
Tislelizumab + TKIs (26)	2022	44	–	–	47.7%[‡]	34.1% (3 pCR cases)
PD-1 monoclonal antibody + TKIs (27)	2021	63	1 Ib case/1 IIa case/1 IIb case/7 IIIa cases	3.2 [2.4–8.3] months	–	15.9% (6 pCR cases)
Nivolumab + cabozantinib (28)	2020	1	–	8 weeks	–	pCR
PD-1 monoclonal antibody + lenvatinib (29)	2020	33	–	–	45.5%[†]	42.4%
Large molecule anti-angiogenesis + immune checkpoint inhibitors						
Atezolizumab + bevacizumab (30)	2023	156	–	–	32%[‡]	10.9%
Atezolizumab + bevacizumab (31)	2023	1	–	15 cycles	–	MPR
Sintilimab + bevacizumab biosimilar (32)	2022	30	13 cases with stage B	4.1 [2.0–5.7] months	23.3%[‡]	43.3%
Atezolizumab + bevacizumab (33)	2022	1	1 case with stage C	7 cycles	–	pCR
Atezolizumab + bevacizumab (34)	2021	32	–	–	–	19%

[–], data not traceable; [†], assessment based on modified RECIST criteria; [‡], assessment based on RECIST v1.1; [§], includes not only radical resection but also cases involving radical radiofrequency or radical transarterial chemoembolization. BCLC, Barcelona Clinic Liver Cancer staging system; CNLC, China Liver Cancer Staging; HCC, hepatocellular carcinoma; MPR, major pathological response (defined as less than 50% residual tumor); pCR, pathological complete response; PD-1, programmed death protein 1; TKIs, tyrosine kinase inhibitors.

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TACE alone (48.8% compared to 9.5%, P<0.001), where 56.1% of the patients had tumor diameter ≥10 cm (42). Additionally, TACE combined with PVE not only reduces tumor burden but also promotes rapid compensatory hypertrophy of the FLR, creating conditions for conversion to resection (43).

A multicenter, retrospective, real-world study conducted in 2023 in China (CHANCE001) showed that for unresectable HCC, the efficacy of TACE combined with TKIs and PD-1/PD-L1 inhibitors was superior to TACE alone, significantly improving the ORR (60.1% compared to 32.0%, P<0.001, based on modified RECIST) (44). The LAUNCH study (NCT03905967) demonstrated that lenvatinib combined with TACE improved outcomes in patients with advanced HCC, with 26 out of 170 patients (15.3%) converting to surgical resection, including 2 achieving pathological complete response (pCR) (45). Another multicenter retrospective study included 181 patients with unresectable HCC treated with TACE combined with lenvatinib and PD-1 inhibitors, 77 of whom successfully converted, with a median conversion time of 4.3 months (46). A prospective, single-arm phase II trial reported at 2023 ASCO showed that TACE combined with donafenib and camrelizumab for unresectable HCC achieved an ORR of 81.3% (based on modified RECIST) and a conversion rate of 80% (47). It is important to consider liver function protection when combining TACE with systemic antitumor therapy, prioritizing drugs that have minimal impact on liver function.

Statement 5: precision TACE is recommended to enhance the ORR while controlling the number of TACE sessions to prevent liver function damage caused by multiple TACE procedures. The combination of TACE with other locoregional or systemic antitumor therapies can further increase the conversion rate (Level of evidence 2, Recommendation A).

HAIC

In recent years, HAIC based on the FOLFOX regimen (fluorouracil + oxaliplatin + leucovorin) has shown favorable outcomes in patients with intermediate-to-advanced HCC in China. Lyu *et al.* (41) reported results from a phase III trial on advanced HCC, showing that HAIC demonstrated a higher efficacy rate (ORR: 31.5% *vs.* 1.5%, P<0.001, based on RECIST 1.1) and conversion rate (12.3% *vs.* 0.8%) compared to sorafenib. Another phase III trial in China involving 315 patients with unresectable HCC (intrahepatic tumor with diameter ≥7 cm) showed that the HAIC group had a higher ORR (46% *vs.* 18%, P<0.001, based on RECIST 1.1) and conversion resection (24% *vs.* 12%, P=0.002) compared to the TACE group (48). Additionally, combining HAIC with systemic antitumor therapy can further enhance efficacy. An RCT involving HCC patients with portal vein tumor thrombus (PVTT) showed that the combination of HAIC and sorafenib improved OS, progression-free survival (PFS), and ORR (40.8% *vs.* 2.5%, P<0.001, based on RECIST 1.1), with a higher surgical conversion success rate compared to the sorafenib monotherapy group (12.8% *vs.* 0.8%, P<0.001) (49). Another phase II trial involving patients with Vp3/Vp4 PVTT showed that HAIC combined with sorafenib achieved a higher ORR (41% *vs.* 3%, P<0.001, based on RECIST 1.1) (50). Clinical studies incorporating HAIC with targeted therapies and ICIs are increasingly reported. A retrospective study showed that HAIC combined with lenvatinib and toripalimab achieved a higher ORR (59.2% *vs.* 9.3%, P<0.001, based on RECIST 1.1) and a higher conversion rate (12.7% *vs.* 0) compared to lenvatinib alone (51). Updates from ASCO 2022 on two triplet therapies also showed favorable outcomes. One phase II trial reported an ORR of 70.96% (based on RECIST 1.1) for HAIC combined with apatinib and camrelizumab (52). Another phase II study reported an ORR of 66.7% (based on modified RECIST) and a conversion success rate of 66.7% for HAIC combined with sintilimab and a bevacizumab biosimilar (IBI305) (53). Two updated retrospective analyses from ASCO 2023 showed that the combination of lenvatinib and camrelizumab with HAIC achieved an ORR of 56.3% (based on RECIST 1.1) and a conversion rate of 31.25% (54). The combination of lenvatinib and tislelizumab with HAIC achieved an ORR of 94.4% (based on modified RECIST) and a conversion rate of 38.9% (55). A Phase II trial reported at ESMO 2023 demonstrated that donafenib combined with sintilimab HAIC achieved an ORR of 80.6% (based on modified RECIST) and a conversion rate of 58.3% (56). These results may be specific to patients with large HCC, diffuse HCC, or those with PVTT. In other cases with different tumor burdens, TACE remains the main treatment option.

Statement 6: for unresectable HCC patients with high tumor burden or PVTT, HAIC demonstrates a high ORR and conversion rate (Level of evidence 2, Recommendation B). Currently, the recommended chemotherapy regimen for HAIC is FOLFOX, which typically requires completion of ≥4 consecutive cycles. Combining HAIC with targeted therapy and/or immunotherapy can

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further enhance the conversion rate (Level of evidence 2, Recommendation B).

Application of radiation therapy in oncological conversion

HCC is sensitive to radiation therapy, and moderate doses can achieve good tumor response rates. For patients who are not suitable for surgical resection, conversion to surgical resectability following radiation therapy can result in prolonged survival. A retrospective analysis in 2014 by Lee *et al.* in South Korea (57), which included 264 HCC patients undergoing concurrent chemoradiotherapy, showed that intrahepatic lesions receiving approximately 45 Gy in 25 fractions combined with 5-fluorouracil (FU) and cisplatin resulted in 18 patients gaining the opportunity for surgical resection, with a median OS of 40 months.

A retrospective study included patients with locally advanced, initially unresectable HCC who received radiation therapy combined with HAIC. Among these, 41 patients (16.9%) achieved successful conversion and subsequently underwent surgical resection, with the surgical and non-surgical groups having 5-year OS rates of 49.6% and 9.8% respectively (P<0.001) (58). A recent study in South Korea collected data from 637 patients with locally advanced HCC treated with radiation therapy concurrent with HAIC between 2005 and 2016. The study categorized patients based on radiation doses into <72 and ≥72 Gy groups, showing the conversion rates of 12% and 20% respectively (P=0.03); post-conversion, the median OS for patients who underwent surgery was significantly extended to 104 months compared to 11 months for those who did not undergo surgery (P<0.001) (59).

Radiation therapy can be combined with targeted therapies and ICIs for conversion therapy. A recent prospective study conducted by Queen Mary Hospital at the University of Hong Kong explored the outcomes of unresectable HCC treated with TACE + stereotactic body radiotherapy (SBRT) + PD-L1 inhibitor (avelumab) in a sequential regimen. 12% (4/33) of patients achieved radical treatment (2 for surgery, 2 for ablation) after the triple therapy, with a 24-month OS rate of 100% (60). Additionally, a propensity score-matched analysis showed that patients treated with SBRT combined with lenvatinib had significantly higher median OS (16.8 *vs.* 11.0 months, P=0.043) and ORR (54.29% *vs.* 22.86%, P=0.007, based on modified RECIST), suggesting that this combination could become a viable option for conversion treatment (61).

Statement 7: combining radiation therapy with other locoregional or systemic antitumor therapies can further enhance the success rate of conversion therapy (Level of evidence 3, Recommendation B).

The necessity, timing and considerations for conversion therapy

The necessity of surgical resection after successful conversion

The significance of conversion therapy is to provide patients with the opportunity for curative resection, thereby achieving longer tumor-free survival and OS, and reducing the duration of maintenance therapy. Most current studies on conversion therapy primarily focus on short-term benefits such as resection rates and postoperative recurrence rates, with few studies using long-term survival as a primary endpoint. Several retrospective studies have shown that survival rates after conversion resection are higher than those from palliative treatments like TACE. For instance, Fan *et al.* (62) reported that after TACE-induced conversion resection, the 1-, 3-, and 5-year OS rates were 80%, 65%, and 56%, respectively. Kulik *et al.* (63) found that patients converted by transcatheter arterial radioembolization (TARE) had 1-, 2-, and 3-year OS rates of 84%, 54%, and 27%, respectively. Lewandowski *et al.* (64) noted that the postoperative tumor-free survival times for patients treated with TACE and TARE were 7.1 and 17.7 months, respectively. Shindoh *et al.* (65) and Kaneko *et al.* (66) observed that patients receiving TKIs for conversion resection had significantly prolonged tumor-free survival and OS. Zhu *et al.* (27) found that patients converted by TKIs combined with ICIs had no tumor recurrence at a median follow-up of 11 months, with four patients discontinuing medication. In another study by Zhu *et al.* (24), long-term survival following TKI plus ICI-induced conversion resection was monitored, showing that the median OS for the surgery group was not reached compared to 15.9 months for the non-surgery group (P<0.001), with 1- and 2-year OS rates of 95.8% and 53.2%, and 95.8% and 35.8% respectively; 14 out of 24 patients who underwent surgery (60.9%) discontinued medication. However, these retrospective studies may have selection biases, and the lack of uniform definitions for unresectable HCC and the standards for surgical resection affect the comparability of survival data. Zhang *et al.* (23) recently published results from a prospective phase II trial

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showing that after lenvatinib combined with PD-1 inhibitor conversion therapy, the 12-month recurrence-free survival (RFS) rate was 47.6%, with a median OS of 31.4 months [95% confidence interval (CI) from 12.6 months to not reached].

In patients undergoing conversion therapy, potent systemic antitumor therapies have been found to achieve pCR (27,29,67,68). Currently, there is no evidence supporting long-term survival data for HCC patients achieving pCR with non-surgical treatment continued. However, without surgical treatment, it is also impossible to confirm that patients have reached pCR.

For patients who achieve radiographic response after treatment, the necessity of surgery remains undetermined. Current studies indicate that most patients with radiographic response, even with continued medication, progress within approximately 1 to 1.5 years. For instance, the median duration of response (DoR) for patients treated with lenvatinib combined with pembrolizumab is 16.6 months (69), and the median DoR for patients treated with bevacizumab combined with atezolizumab is 18.1 months (5). Drawing from experiences with colorectal cancer liver metastases, even lesions that disappear on imaging after chemotherapy (radiographic complete response) have a >50% recurrence rate upon continued follow-up (70). Therefore, surgical resection is anticipated to provide patients with longer tumor-free survival and OS. Additionally, conversional resection has significant implications for reducing drug exposure and associated adverse reactions. Although the degree of remission might be higher with targeted therapy combined with ICIs, whether inactive lesions that disappear on imaging or lose contrast enhancement still require surgical resection needs to be addressed by prospective controlled studies. Concurrently, the risks and trauma of surgical resection itself must also be evaluated.

Statement 8: surgical resection is a crucial method for achieving long-term survival after successful conversion therapy. Surgical resection not only eliminates potential residual tumor cells but also provides pathological insights that can guide subsequent adjuvant therapy (Level of evidence 3, Recommendation A).

Timing of surgical resection after successful conversion

Choosing the right timing for surgery based on tumor response

Many scholars believe that for patients initially unsuitable for surgical resection, surgery should be performed as soon as they meet the surgical criteria. However, studies show that tumor-free survival post-conversional resection correlates with the degree of pathological response, with patients exhibiting pathological remission experiencing longer postoperative tumor-free survival (67,71). Therefore, the indicators of successful conversion are not only the suitability for surgery but also the degree of tumor response, which relates more closely to postoperative recurrence and long-term survival. Achieving objective response (tumor reduction or downstaging) or maintaining lesion stability for a period (e.g., 3–4 months) may be prerequisites for obtaining better oncological outcomes after resection.

Statement 9: The indicators of successful conversion are not only the suitability for surgery but also the degree of tumor remission, which is more closely related to postoperative recurrence and long-term survival (Level of evidence 3, Recommendation B).

Choosing the right timing for surgery based on safety

The timing of surgery in conversion therapy should also consider the safety of the surgery. The appropriate timing for surgery varies depending on the preoperative treatment. There is little research specifying the duration for which medication should be discontinued before surgery for systemic antitumor therapy. Small molecule targeted drugs are taken continuously, and currently, there is no clear guidance on when to discontinue them before surgery. According to existing literature, continuing medication does not increase the incidence of postoperative complications (72,73), and some case reports suggest that surgery can be performed a week after discontinuing medication (74). Bevacizumab has a half-life of about 20 days, and its anti-angiogenic effects may increase surgical bleeding and affect wound healing (75). From the experience of liver resection for colorectal cancer metastases, it is generally recommended to discontinue bevacizumab for more than 4 weeks before surgery to ensure safety. ICIs are typically administered on a cyclical basis, and existing literature suggests that HCC conversion surgery should be performed within 4 weeks after the last treatment cycle (76,77). If adverse reactions occur during targeted therapy or ICI treatment, surgery should proceed only after the drug is discontinued and adverse reactions have returned to Grade I or normal. Additionally, immunogenic hepatitis may increase the risk of surgical mortality. Thus, for patients undergoing conversion therapy with PD-1/

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PD-L1 inhibitors, it is crucial to assess the presence of immune-related hepatitis during surgical safety evaluations. For assessing immune-related hepatitis, in addition to routine markers of hepatocellular damage like alanine aminotransferase (ALT) and aspartate aminotransferase (AST), a liver biopsy may be necessary to observe inflammatory and lymphocytic infiltration, as well as hepatocellular necrosis. Currently, there is limited research on the impact of anti-angiogenic drugs combined with PD-1/PD-L1 inhibitors on postoperative liver function and surgical safety, and more data are needed to establish an appropriate preoperative assessment strategy. Shen *et al.* (78) found that patients undergoing resection after conversion therapy with PD-1 inhibitors combined with TKIs experienced more intraoperative bleeding and longer hospital stays compared to those undergoing direct surgery, though the incidence of complications at 30 days and mortality rates at 90 days post-operation were similar. Luo *et al.* (79) concluded similarly in a study using locoregional therapy combined with TKIs and PD-1 inhibitors as conversion therapy. Therefore, patients planned for post-conversion resection require precise preoperative assessment and perioperative management to ensure the safety of the surgery.

Previous studies have found that preoperative TACE can induce liver inflammation, increasing intraoperative bleeding and the difficulty of surgical operations (80). However, many scholars point out that when there is a sufficient interval between the last TACE and surgery, the impact of TACE on surgery is minimal, and it does not significantly affect the incidence of perioperative complications or mortality. It is recommended that the interval between the last preoperative TACE and surgery be at least 4 weeks (with a median interval of 6 weeks) (81,82). Although some patients may experience perihepatic inflammatory adhesions within weeks after TACE, these do not significantly negatively affect surgical operations or short-term surgical outcomes. In patients undergoing TACE combined with PVE as conversion therapy, it is recommended to assess the FLR every 2 weeks after PVE, and surgery can be considered when FLR and ICG R15 meet surgical criteria (83). Impaired liver function is common during TACE treatment, and surgery should be performed after recovery of liver function. The interval between the radiotherapy for PVTT in the main trunk and hepatectomy is generally 8 weeks, which can minimize liver function impairment, intraoperative bleeding, and postoperative liver failure (84).

Statement 10: the safety of surgery is an important aspect of the assessment before conversional resection. It is necessary not only to evaluate the routine safety checks required for hepatectomy but also to assess the potential impact of prior conversion therapy on the liver. Before surgery, it is recommended to discontinue small molecule targeted drugs for more than 1 week, PD-1/PD-L1 inhibitors for more than 2 weeks, and bevacizumab for more than 4 weeks; if undergoing TACE, surgery should be performed 4 weeks after the last TACE; the interval between the radiotherapy for PVTT in the main trunk and hepatectomy is typically 8 weeks (Level of evidence 4, Recommendation B).

Postoperative treatment after successful conversion resection

For patients who meet the criteria for surgical resection after conversion therapy, the preoperative conversion therapy regimen has been proven to be effective, so it is preferable to continue using the original conversion therapy regimen for adjuvant therapy postoperatively. However, the use of postoperative adjuvant therapy should balance efficacy and safety. Combining literature on HCC conversion therapy and other tumor conversion experiences, if the conversion therapy involves combinations of drugs, such as targeted therapy & immunotherapy or dual ICIs combinations, the choice should be based on the patient's physical condition, adverse reactions, and treatment tolerance, considering either the original regimen or part of the drugs from the original regimen. If the conversion therapy regimen involves drug combinations with locoregional therapy, since the target lesion disappears postoperatively, consider 1–2 preventive TACE/HAIC postoperatively followed by drug maintenance (79,85), or just use the drugs from the original conversion regimen and adjust according to the he patient's condition.

As for the duration of adjuvant therapy after successful resection, there is currently insufficient data. Based on research on adjuvant therapy after resectable HCC and the currently common conversion therapy regimen with recurrence free survival data, it is suggested that postoperative adjuvant therapy lasts 6–12 months. Additionally, it is recommended to follow up every 3 months, closely monitoring changes in tumor markers and imaging. If two consecutive imaging results show no tumor metastasis or recurrence, and tumor markers remain normal for 3 months without an upward trend, consideration

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should be given to discontinuing medication. During adjuvant therapy, closely monitor adverse reactions; if severe adverse reactions occur or the patient cannot tolerate them, consider reducing or discontinuing the medication. For patients whose tumor specimens achieve pCR, consider shortening the postoperative adjuvant therapy duration.

Due to insufficient evidence for conversion therapy after HCC conversion resection, it is encouraged to undertake clinical studies beyond current guidelines and consensus, such as the ongoing TALENTOP study.

Statement 11: after successful conversion therapy for HCC, the original conversion therapy regimen can be used for adjuvant therapy, with adjustments made based on the patient's physical condition, adverse reactions, and tolerance. It is recommended that postoperative adjuvant therapy continue for 6 to 12 months, with follow-up every 3 months. If two consecutive imaging assessments show no evidence of tumor recurrence or metastasis, and tumor markers remain normal without an upward trend for 3 consecutive months, consideration may be given to discontinuing the medication (Level of evidence 3, Recommendation A).

Subsequent treatment for patients who do not achieve successful conversion

For patients with intermediate-to-advanced HCC, whether or not they can undergo surgical resection after conversion therapy, the primary goal of treatment is to extend the patient's survival while also focusing on the quality of life. Whether surgery can be achieved after conversion therapy depends not only on tumor reduction but also on the tumor's location, the patient's liver function, and physical condition. Thus, for patients who do not achieve conversional resection, treatment should be personalized based on the patient's specific situation under the guidance of relevant guidelines, using high-level evidence as the basis.

For patients who fail first-line treatment (including tumor progression and treatment intolerance), it is necessary to switch to second-line treatment options promptly, or choose first-line treatment drugs that have not been used before, or combine other treatment modalities (such as locoregional therapy), or consider participating in clinical trials. For example, if first-line treatment with PD-1/PD-L1 inhibitors combined with bevacizumab fails, subsequent treatment options might include TKIs such as lenvatinib or sorafenib (86,87). For patients who fail first-line treatment with PD-1/PD-L1 inhibitors, combining treatment with CTLA-4 antibodies may be considered (88). If first-line treatment involves TKIs, like sorafenib or lenvatinib, second-line treatment might involve switching to regorafenib or apatinib (89-91), or switching to ICIs treatment, including camrelizumab, tislelizumab, pembrolizumab, etc. (92-94). For patients resistant to or progressing after TACE, combining systemic antitumor therapy or switching to a standalone systemic antitumor therapy mode is recommended. For patients with PR or stable disease (SD) but still no opportunity for surgical resection, maintaining the current treatment regimen is appropriate. For patients with local progression, switching to second-line treatment as well as combining with locoregional therapies should be considered. Additionally, for patients with intermediate-to-advanced HCC experiencing disease progression, it is also important to pay attention to follow-up and treatment of underlying diseases. For example, close monitoring of hepatitis B virus deoxyribonucleic acid (HBV-DNA) levels is essential; if there is an increase in HBV-DNA levels, antiviral therapy should be adjusted to more effective HBV medications.

Statement 12: subsequent treatment for patients who do not achieve successful conversion should consider a comprehensive decision-making process that includes underlying liver disease, prior treatment methods, tumor progression characteristics, and patient preferences. Based on the pattern of disease progression, timely switching to a second-line treatment regimen is recommended. This may include using previously unused first-line drugs, combining with other treatment modalities, or participating in suitable clinical trials (Level of evidence 2, Recommendation A).

Perioperative treatment

Neoadjuvant therapy

Value of neoadjuvant therapy

Neoadjuvant therapy involves systemic or locoregional anti-cancer treatments administered before surgery to patients eligible for resection but at high risk of recurrence and metastasis. Its efficacy is established in various cancers including malignant melanoma, lung cancer, and breast cancer. Its significance includes: (I) Early management of occult lesions and micrometastases reduces postoperative recurrence risk; (II) improved prognosis in patients with locally advanced disease; (III) enhanced preoperative physical condition and treatment tolerance; (IV) since the

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tumor is not yet resected, neoadjuvant therapy can induce the release of more tumor-associated antigens, potentially improving the efficacy of immunotherapy; (V) providing insights into tumor drug sensitivity and biological behavior, aiding in the selection of postoperative adjuvant therapy; (VI) facilitating the collection and analysis of pre- and post-treatment tissue and blood samples, advancing research on biomarkers and resistance mechanisms.

Populations for neoadjuvant therapy

Neoadjuvant therapy is appropriate for patients in CNLC stages Ib (equivalent to BCLC A) to IIa (equivalent to BCLC B), and some in stages IIb (equivalent to BCLC B) and IIIa (equivalent to BCLC C), suitable for resection but with high postoperative risks of recurrence and metastasis. Factors assessed preoperatively include macroscopic tumor thrombus, tumor diameter >5 cm, multiple tumors, involvement of adjacent organs, high preoperative alpha-fetoprotein (AFP) levels, and high preoperative serum HBV-DNA load.

Neoadjuvant therapy strategies

While aiming to eradicate micrometastases and reduce postoperative recurrence risk, neoadjuvant therapy poses risks such as potential adverse reactions that could delay surgery, increase postoperative complications, preclude postoperative adjuvant therapy, or result in the loss of a surgical opportunity due to tumor progression. Therefore, careful selection of suitable patients and appropriate treatment plans based on therapy goals is crucial. Treatment regimen selection should consider not only ORR but also higher disease control rates to prevent loss of surgical opportunity due to disease progression, favoring safer treatment modalities with fewer adverse reactions. The purpose of neoadjuvant therapy is to improve RFS without compromising the surgical plan, typically lasting 1.5 to 3 months, not exceeding 4 months.

Most studies on neoadjuvant therapy for HCC are small-scale phase II clinical trials with limited high-level evidence. A multicenter, prospective phase III RCT from Sun Yat-sen University Cancer Center showed that HAIC neoadjuvant therapy for BCLC A/B stage patients exceeding Milan criteria achieved a 63.6% ORR (based on modified RECIST) with significant improvements in OS and PFS (3-year OS rate: 63.5% *vs.* 46.3%, P=0.016; median PFS: 14.1 *vs.* 8.9 months, P=0.017) (95). Wei *et al.* (96) compared the efficacy of preoperative radiotherapy combined with surgical treatment versus surgery alone in patients with CNLC stage IIIa (equivalent to BCLC C) HCC. In the radiotherapy group, 17 cases (20.7%) experienced a downgrade in PVTT from Cheng's type III to type II or from type II to type I. The combination of preoperative radiotherapy and surgical treatment significantly extended the OS compared to surgery alone (24 months, 27.4% *vs.* 9.4%, P<0.001). Hokkaido University in Japan categorized HCC patients with PVTT in the main trunk or first branch into a group receiving radiotherapy combined with surgery and a surgery-only group. The radiotherapy regimen was 30–36 Gy administered over 10–12 sessions. Postoperative pathological examinations indicated that 83.3% (5/6) of patients with PVTT in the main trunk achieved a pCR; the 5-year OS rate was 34.8% for the combined treatment group compared to 13.1% for the surgery-only group (P=0.0359) (97). A prospective phase II trial at the Cancer Hospital Chinese Academy of Medical Sciences revealed that 38 patients with central-type HCC receiving intensity-modulated conformal radiotherapy as neoadjuvant therapy had 16 cases (42.1%) with PR and 22 cases (57.9%) with SD, with no patients showing disease progression. Thirteen patients (34.2%) achieved a major pathological response (MPR), including 5 cases (13.2%) with pCR. The 1-, 3-, and 5-year OS rates were 94.6%, 75.4%, and 69.1% respectively; the PFS rates were 70.3%, 54.1%, and 41.0% (98). In recent years, the efficacy of systemic antitumor therapy has improved, and there is active exploration of targeted therapies, immune checkpoint ICIs, or combined targeted therapy & immunotherapy regimens in neoadjuvant settings. However, most of these studies are small-scale phase II exploratory trials, with primary endpoints often focused on pathological response rates rather than OS. More prospective, multicenter clinical studies are needed for validation. A recent sequential phase II/III study from Zhongshan Hospital Fudan University, using a combination of camrelizumab and apatinib for perioperative treatment in resectable CNLC stage Ib–IIIa (equivalent to BCLC A–C) HCC, showed that 40% of patients in the neoadjuvant therapy group achieved MPR (99).

Statement 13: neoadjuvant therapy is defined as the preoperative administration of systemic anti-tumor or locoregional therapy to HCC patients suitable for surgical resection but with a high risk of recurrence or metastasis, aimed at reducing the risk of postoperative recurrence and metastasis. However, it carries risks and should be strictly reserved for appropriate candidates, with treatment plans carefully tailored to therapy goals (Level of evidence 2, Recommendation B).

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Postoperative adjuvant therapy

Populations for postoperative adjuvant therapy

Postoperative adjuvant therapy is critical for reducing the risk of recurrence and improving long-term survival. Unlike neoadjuvant therapy, it allows for more precise selection of treatment populations and personalized treatment plans based on postoperative pathology and molecular typing, without delaying surgical risks. It primarily benefits patients suitable for resection with high risks of recurrence and metastasis. Common factors assessed postoperatively include tumor rupture, tumor diameter >5 cm, multiple tumors, MVI, major vascular invasion, positive or narrow margins, and Edmondson grade III–IV tumors (100-102).

Postoperative adjuvant therapy strategies

For patients with a high risk of recurrence and metastasis after surgery, there is currently no international standard adjuvant therapy. Results from RCTs indicate that postoperative TACE can effectively reduce recurrence and prolong survival (103,104). A multicenter phase III RCT conducted by Sun Yat-sen University Cancer Center demonstrated that FOLFOX-HAIC can reduce the recurrence risk and improve survival in HCC patients with MVI (105). Another prospective multicenter phase III RCT showed that Chinese herbal Huai Er granule can reduce postoperative recurrence and prolong survival (106). A single-arm, multicenter, prospective study indicated that postoperative adjuvant therapy with lenvatinib alone achieved a median RFS of 19.33 months (107). A prospective phase II trial from Cancer Hospital Chinese Academy of Medical Sciences showed that for HCC patients with narrow margins (<1 cm), adjuvant radiotherapy postoperatively resulted in a 5-year OS rate of 72.2% and a disease-free survival rate of 51.6% (108). A prospective multicenter phase III RCT from South Korea revealed that activated cytokine-induced killer (CIK) cell therapy could significantly prolong median PFS in patients undergoing surgical resection, RFA, or ethanol injection, with median RFS of 44 months in the treatment group compared to 30 months in the control group (109). Additionally, for HCC patients with HBV infection, antiviral treatment with nucleoside analogues helps reduce postoperative recurrence and should be taken long-term (110). For patients with viral hepatitis-related HCC, postoperative adjuvant pegylated interferon can improve OS and RFS without causing severe adverse reactions (111). Recent years have seen ongoing in-depth research into systemic anti-cancer therapy in HCC adjuvant therapy, with several phase III clinical trials actively exploring options, such as the IMbrave 050 study (atezolizumab combined with bevacizumab versus active surveillance), JUPIER-04 (toripalimab versus placebo), EMERALD-2 (durvalumab combined with bevacizumab versus placebo), KEYNOTE 937 (pembrolizumab versus placebo), CheckMate-9DX (nivolumab versus placebo), and SHR-1210-III-325 (camrelizumab combined with apatinib versus active surveillance). Among these, the IMbrave 050 study showed that atezolizumab combined with bevacizumab can reduce the risk of recurrence by 28% (112). However, although this combination initially improved 1-year RFS compared to active surveillance (78% *vs.* 65%) in a population mainly consisting of BCLC stage A patients, the advantage was not maintained with longer follow-up (113).

Statement 14: for HCC patients suitable for surgical resection and at high risk of postoperative recurrence and metastasis, adjuvant therapies such as antiviral therapy, TACE, HAIC, targeted therapy combined with ICIs, radiation therapy, and Chinese herbal Huai Er granule can be employed to reduce postoperative recurrence and improve long-term survival rates (Level of evidence 1, Recommendation B).

Assessment of lesions during conversion therapy and perioperative treatment

Pathological assessment

After conversion and perioperative treatment, surgical specimens are generally independently evaluated by a pathologist. The assessment includes the histological morphology related to the treatment and the proportion of residual tumor. A gross examination of the main tumor body is conducted to preliminarily assess the proportion of residual active tumor, which is described in the macroscopic examination report. Hematoxylin and eosin (H&E) stained slides are microscopically examined to assess the inactive tumor areas in the tumor bed, including necrosis, fibrosis, hemorrhage, foamy cell reaction, cholesterol crystal phenomena, tumor-infiltrating lymphocytes, and tertiary lymphoid structures, and the proportion of residual active tumor areas is calculated. The proportion of residual tumor needs to be assessed by percentage for the regression bed (pathological remission) area, with 5% as the minimum increment for assessment; the original tumor body minus the sum of various treatment reaction proportions equals the residual tumor proportion, i.e., residual tumor

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proportion (%) = residual tumor area (1 – regression bed) / total tumor bed area × 100%.

Immunologically related pathological response criteria (irPRC) are applied in the evaluation of final surgically cleared lymph nodes. Compared to the current American Joint Committee on Cancer (AJCC) staging applied to lymph node components and the residual cancer burden calculator used for neoadjuvant therapy, irPRC provides higher resolution and offers a standard for measuring whether tumor cells can achieve any immune-mediated clearance through neoadjuvant therapy.

When sampling resected specimens after conversion therapy and perioperative treatment, based on imaging and gross findings, extensive necrosis should be accurately described in terms of scope and percentage during gross description and photographed. For small HCC ≤3 cm in maximum diameter, entire sampling is required; for tumors >3 cm, sampling should be done at intervals of 0.5–1 cm through continuous sectioning, recording the percentage of necrosis visible in each cut surface, with representative sections fully sampled according to the seven-point sampling scheme, and selective sampling of other levels with photography. If feasible, large embedding cassettes and large slides should be used. Additionally, post-treatment lymph nodes and cancer thrombi should be fully sampled, and multi-nodular lesions should be sampled according to the single nodular lesion sampling model.

In clinical evaluations of HCC conversion and perioperative treatment regimens, survival rates and ORR data require patient follow-up over several years. In contrast, pathological response assessment can provide more accurate early feedback within weeks or months, and also determine the necessity for adjuvant therapy post-resection, thus offering a significant application advantage. Since there is no unified grading standard for pathological assessment of neoadjuvant therapy in HCC, based on neoadjuvant pathological assessments for other solid tumors such as non-small cell lung cancer and malignant melanoma, patients can currently be categorized into MPR or pCR based on pathological remission.

MPR is defined as the reduction of viable tumor cells to a clinically significant threshold, commonly defined in lung cancer studies as a reduction of residual tumor cells in the tumor bed to ≤10% (114), which correlates with studies on the extent of tumor necrosis following TACE treatment and prognosis in HCC (39). However, the threshold for MPR varies among different histological types (115), and thus, the definition for MPR in HCC has not been established but should generally represent at least a 50% reduction. It is recommended to expand the sampling scope of tumors initially diagnosed as MPR to clarify. In cases where primary liver tumors have few or no viable tumor cells, but viable metastatic foci are present in lymph nodes/cancer thrombi (ypT0, N1, 2, or 3), these can also be classified as MPR, although their prognostic and therapeutic significance remains unclear (114).

pCR is defined as no residual tumor tissue observed in the original tumor bed, vascular cancer thrombi, and lymph nodes, indicating that no viable tumor cells are found in the completely assessed resected specimen, including all sampled areas of lymph nodes, cancer thrombi, and distant metastases, after reviewing all sections. Preliminary clinical evidence indicates that patients achieving MPR or pCR have better postoperative survival outcomes than those who do not achieve MPR or pCR, although more evidence is needed (67). Additionally, how to predict MPR or pCR through imaging or other clinical markers remains an urgent issue to address. Studies by Huang *et al.* (116) show that imaging complete response is not a necessary condition for pCR; the measurement method according to the modified RECIST criteria, where the preoperative tumor enhancement area reduces by ≥80% compared to baseline, can predict pCR well, and combining changes in AFP can better predict pCR.

Statement 15: MPR is defined as a reduction in the proportion of viable tumor cells to a clinically significant threshold. Specimens classified as achieving a pCR should be comprehensively sampled and cautiously evaluated. The topic of predicting pathological remission of tumors through monitoring dynamic changes in imaging and tumor markers requires further research (Level of evidence 3, Recommendation B).

Imaging assessment

Computed tomography (CT) and magnetic resonance imaging (MRI) are standard methods used to assess the efficacy of conversion therapy and perioperative treatment in HCC, select the timing of surgery, and determine the resectability of conversion surgery. MRI, with its high soft tissue resolution and insensitivity to iodine oil, provides greater accuracy in determining the active range of the tumor compared to dynamic enhanced CT scans. Dynamic enhanced MRI, especially using liver-specific contrast agents, is particularly effective in detecting small HCC (117). On the other hand, CT scans, with their high

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spatial resolution, are especially advantageous in precisely calculating the FLR/total liver volume ratio and simulating liver resections in patients with large HCC, thus reducing the surgical risks associated with conversion therapy. It is recommended to use the same imaging method for baseline and follow-up assessments to minimize systematic errors.

Compared to the traditional RECIST 1.1 criteria, the modified RECIST criteria are more suitable for assessing the response of HCC to treatment (2). Seymour *et al.* (118) have shown that the modified RECIST criteria distinguish the objective response to targeted therapy in HCC better than the RECIST 1.1 standards, and the results correlate with patient survival. While the modified RECIST criteria correlate tumor response with pathological changes, imaging-enhanced tissue cannot differentiate between tumor tissue and inflammatory responses, limiting the ability to predict the degree of pathological remission. Combining serological markers such as AFP, abnormal prothrombin, and a combination of 7 microRNAs (miRNAs) may further improve the accuracy of predicting pCR (116). The complexity of treatment responses in HCC has increased with immunotherapy, necessitating a comprehensive assessment combining immune RECIST (iRECIST) standards. Advanced techniques integrating medical engineering, such as radiomics and artificial intelligence, offer potential for in-depth analysis of imaging data to extract biologically relevant information, enabling personalized predictions of therapeutic response and prognosis in HCC. A multicenter study has shown that pre-treatment radiomics based on dynamic enhanced MRI correlates with OS and PFS, and can be used to predict the therapeutic efficacy of combined target therapy & immunotherapy in HCC patients (119).

Statement 16: dynamic enhanced MRI and the modified RECIST are recommended as the preferred imaging assessment methods for evaluating the efficacy of conversion therapy and perioperative treatment in HCC (Level of evidence 3, Recommendation A).

Management of complications during conversion therapy and perioperative treatment

Before implementing combined therapy regimens for conversion therapy and perioperative treatment, a clinical examination of the patient should be conducted, and baseline patient conditions assessed. During treatment, close monitoring of adverse reactions and complications is crucial to ensure timely prevention, detection, and management. Common adverse reactions to targeted therapies include hand-foot skin reactions, hypertension, diarrhea, decreased appetite, weight loss, proteinuria, rash, liver function abnormalities, thrombocytopenia, and hypothyroidism (89,91,120,121). When using ICIs, the most commonly affected organs or systems by immune-related adverse events (irAEs) include the skin, colon, liver, and endocrine systems. Life-threatening irAEs, although less common, include interstitial pneumonia and immune-mediated myocarditis. Different ICIs have varying spectrums of irAEs: severe irAEs associated with CTLA-4 antibodies are primarily colitis, while serious irAEs with PD-1/PD-L1 inhibitors typically include interstitial pneumonia, hepatitis, and neurotoxicity (122). The combination of targeted drugs and ICIs may increase the incidence of certain adverse reactions, such as skin toxicity, diarrhea, fatigue, and hypothyroidism (123-125). Common adverse reactions to the combination of ICIs and bevacizumab include hypertension, elevated AST, proteinuria, and fatigue (15). When using bevacizumab, vigilance is needed for potential bleeding, perforation, and thromboembolic events. During treatment, regular monitoring of adverse reactions is recommended.

Management should follow the principles outlined in the "National Cancer Institute Common Terminology Criteria for Adverse Events Version 5.0" (NCI-CTCAE 5.0), "Guidelines of Chinese Society of Clinical Oncology (CSCO) for the management of toxicities immune checkpoint inhibitor-related toxicity (2021)", "Chinese Expert Consensus on the clinical application of molecularly targeted drugs in hepatocellular carcinoma (2022 edition)", and other relevant guidelines and consensus. Timely diagnosis and active treatment of adverse reactions are necessary, with adjustments to drug dosage, temporary suspension, or termination of treatment as appropriate (126-128). Medical centers with the capability should establish an MDT for the diagnosis and treatment of adverse reactions, enabling multidisciplinary management.

Statement 17: during the conversion therapy and perioperative treatment for HCC, close monitoring of adverse reactions and complications is essential for timely prevention, detection, and management (Level of evidence 1, Recommendation A).

Prediction of efficacy and exploration of related mechanisms in conversion therapy and perioperative treatment

The efficacy of conversion therapy is a prerequisite for

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implementing conversion resection, making the prediction of patient responses to conversion therapy particularly important before initiating such treatment. Currently, there is a lack of widely recognized biomarkers to predict the efficacy of targeted therapy, immunotherapy, locoregional therapy, or combined therapies, although some studies provide useful references.

Using clinical features, laboratory and imaging examinations to predict the efficacy of systemic anti-tumor therapy holds clinical value. Reports indicate that the combined score of serum protein induced by vitamin K absence or antagonist-II (PIVKA-II) and metastatic status (129), the ratio of white blood cell (WBC) count to lymphocyte percentage (130), the number of naive $CD8^+$ T cells in peripheral blood (131), and radiomic data from pre-treatment enhanced MRI or CT scans (118,132) can predict the degree of tumor response to treatment with lenvatinib alone or in combination with PD-1/PD-L1 inhibitors.

Currently, the combination of bevacizumab and PD-L1 inhibitors has become the preferred first-line treatment for advanced or unresectable hepatocellular carcinoma. Reports suggest that peripheral blood cell profiling can help predict the efficacy of atezolizumab combined with bevacizumab. Although results vary, generally, patients with a higher baseline or early-treatment ratio of neutrophils to lymphocytes have lower tumor response rates, higher disease progression rates, and shorter PFS and OS (133-138). Additionally, exploratory biomarker studies are noteworthy. Studies have reported that the expression of CD274 (PD-L1), effector T-cell markers, and the density of $CD8^+$ T cells in tumor tissue are associated with a favorable prognosis; in contrast, the ratio of regulatory T cells to effector T cells, the levels of glypican-3 (GPC3), and AFP are associated with a poor prognosis post-treatment (139). Patients with higher serum concentrations of CD137 have longer PFS; those with a higher density of M1 macrophages (CD68 positive, CD163 negative) infiltrating tumor tissue have higher ORR and longer PFS (140). Patients with higher pre-treatment levels of osteopontin and AFP are more likely to be evaluated as having disease progression post-treatment, with shorter PFS and OS (141). However, these biomarkers still lack sufficient clinical validation.

Based on the characteristics of liver tumors, appropriate choices can be made among locoregional therapies such as TACE and HAIC. Currently, there are several clinical criteria based on liver tumor burden to select patients suitable for TACE treatment. It is generally believed that patients with a liver tumor burden exceeding the up-to-7 criteria (the sum of the maximum diameter of the largest tumor and the number of tumors ≤7) or the six-and-twelve criteria (142) benefit less from TACE treatment. There is considerable overlap between the populations suitable for HAIC and those for TACE, but the two can complement each other. The "Chinese expert consensus on hepatic arterial infusion chemotherapy for hepatocellular carcinoma (2021 edition)" can be referenced to make a reasonable choice between TACE and HAIC, thus achieving maximal control of liver tumor lesions.

Apart from tumor sensitivity to treatment regimens, the initial tumor burden also affects the patient's chances of undergoing surgical resection after conversion therapy. A retrospective study shows that patients with the following characteristics have a higher chance of successful conversion after the combination of targeted therapy & immunotherapy and subsequently undergo surgical resection (25): (I) Eastern Cooperative Oncology Group performance status (ECOG PS) score of 0–1, liver function Child-Pugh A; (II) tumor confined to one liver lobe; if there is a tumor in the contralateral lobe, it meets the Milan criteria (single tumor ≤5 cm in diameter; multiple tumors <3, with the largest diameter ≤3 cm); (III) if there is a vascular thrombus, there are no tumor thrombi in the contralateral portal vein branches or extending to the mesenteric vein, and no hepatic vein thrombus extending to the right atrium; (IV) no extrahepatic metastasis. Patients meeting these criteria have about a 50% chance of successful conversion and undergoing surgical resection after treatment with lenvatinib combined with a PD-1 inhibitor; those who do not meet these criteria have ≤10% chance of surgical resection. In clinical practice, selecting patients who meet these criteria for intensive systemic antitumor therapy, combined or not with locoregional treatment, aims to achieve a higher conversion resection rate; patients exceeding these criteria have a relatively lower likelihood of subsequent surgical resection, and treatment goals should be set to prolong survival time based on a thorough assessment of the impact on quality of life.

MDT and its role in conversion and perioperative treatment of hepatocellular carcinoma

The role of MDT in conversion therapy and perioperative treatment

Currently, the modalities used in the conversion therapy and perioperative treatment of HCC include locoregional

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therapies (such as transarterial interventions and radiotherapy) and systemic drug therapies. Each treatment method has its own advantages and disadvantages, with overlapping indications, and the implementation often involves multiple disciplines. The biological behavior of HCC is highly heterogeneous, and there are significant differences in liver disease background and prognostic factors among individuals. At present, there is a lack of clear guidelines and standards regarding the appropriate population, specific methods and plans, timing of surgery, and duration of perioperative treatment for HCC conversion therapy and perioperative treatment. It is necessary to gradually summarize experience in practice and to engage in continuous communication and discussion among different disciplines to form a consensus. Therefore, cooperation and communication within the MDT during the conversion therapy and perioperative treatment are crucial. Consider establishing a relatively fixed MDT team and forming effective and convenient communication channels to ensure that treatment plans can be adjusted in a timely manner according to changes in the patient's condition, maximizing patient benefits.

Coordination and division of labor among MDT professionals during conversion therapy and perioperative treatment

The short-term goals of conversion therapy and perioperative treatment for HCC are to create opportunities for surgical resection, improve surgical outcomes, and prevent and reduce recurrence, with the ultimate goal of achieving high-quality, long-term tumor-free survival for patients. During the conversion therapy and perioperative treatment, all medical personnel should keep these goals in mind and adhere to the "three dos and three don'ts" principles of HCC MDT: focus on the patient, aim for efficacy, base decisions on evidence-based medicine; do not let personal expertise solely dictate the treatment plan, avoid excessive reliance on a single treatment, and do not let economic interests determine the treatment plan. Through MDT, personalized and optimized decision-making is provided for patients with HCC.

The disciplines involved in the conversion therapy and perioperative treatment and their primary roles include:

(I) Surgery: the leading discipline for performing HCC conversion surgery and perioperative treatment. Responsibilities include selecting suitable patients, assessing operability, deciding on the timing of surgery, performing the surgery, and perioperative management.

(II) Interventional radiology and radiation oncology: main disciplines responsible for delivering locoregional therapies such as transarterial interventions or radiotherapy during the conversion therapy and perioperative treatment and providing additional treatments for non-surgical patients after conversion therapy.

(III) Medical oncology: a key discipline in administering systemic antitumor therapy during the conversion therapy and perioperative period, responsible for follow-ups and assessments during the treatment.

(IV) Infectious Diseases or Hepatology: Responsible for the evaluation and treatment of underlying liver diseases (like hepatitis and cirrhosis) and related complications throughout the treatment. They provide recommendations on specific plans and surgical tolerability for conversion therapy and perioperative treatments, and manage long-term follow-up for patients who successfully undergo conversion surgery.

(V) Radiology (including ultrasound): responsible for the radiological diagnosis and evaluation of efficacy. Assists in accurately assessing the tumor status before, during, and after treatment (such as tumor size, active range, number of lesions, tumor-vascular relationships, and preoperative 3D planning), and offers critical advice for selecting patients and timing/methods of surgery.

(VI) Pathology (including molecular testing): responsible for histological diagnosis and efficacy evaluation, assists in identifying causes and degrees of certain adverse reactions (such as immune hepatitis), and helps explore molecular markers that can predict the efficacy of conversion and perioperative treatments and screen appropriate patient populations.

(VII) Other disciplines: in addition to the disciplines routinely involved in the MDT, others such as anesthesiology, intensive care unit (ICU) (for perioperative assessment and management), cardiology, pulmonology, endocrinology, gastroenterology, dermatology (for managing complications and adverse reactions), traditional Chinese medicine, nutrition (for patient conditioning), and nursing (to guide daily care and follow-up) may also participate in the conversion therapy and perioperative treatment

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process.

In summary, MDT collaboration is integral throughout the conversion therapy and perioperative treatment of HCC. The leading or specifically responsible disciplines may vary at different stages of treatment, requiring timely communication and coordination among multiple disciplines based on the progression of the patient's condition. This ensures patient-centered, personalized comprehensive management, aiming to secure high-quality, long-term tumor-free survival for more HCC patients.

Statement 18: MDT collaboration is advocated throughout the entire process of conversion therapy and perioperative treatment for HCC. Medical professionals from multiple disciplines are required to communicate and collaborate timely according to the development of the patient's condition, to achieve patient-centered, individualized comprehensive management. This approach aims to secure high-quality, long-term tumor-free survival for more patients with HCC (Level of evidence 2, Recommendation A).

Summary

Despite the critical role of early screening in the early diagnosis of HCC, 64% of patients in China are diagnosed at intermediate-to-advanced stages, often losing the opportunity for surgical resection, which impacts long-term survival. Advances in surgical techniques (such as ALPPS, PVE), the emergence of new models of systemic anti-tumor therapy (such as anti-angiogenesis combined with immunotherapy), and upgrades in traditional locoregional therapy (HAIC combined with TACE) can significantly improve the ORR. These developments provide important opportunities for conversion therapy in patients unable to undergo surgical resection, allowing some with intermediate-to-advanced stage HCC to achieve long-term survival opportunities. However, how to wisely choose these techniques, determine the timing of surgery after conversion therapy, and select the appropriate population remain key issues in the field of HCC conversion therapy. There is a need for further exploration through more clinical practice and mechanistic studies to improve the prognosis of patients with intermediate-to-advanced HCC.

Another critical factor affecting the long-term survival of HCC patients is the high rate of postoperative recurrence and metastasis. Existing evidence shows that immunotherapy, whether as monotherapy or in combination, can be used as neoadjuvant or adjuvant therapy to reduce the biological behavior of tumors and the risk of postoperative recurrence and metastasis. Adopting strategies such as immunotherapy combined with targeted drugs, immunotherapy as monotherapy, or other combination therapies for neoadjuvant or adjuvant therapy in HCC patients with high risk of recurrence and metastasis is expected to further enhance efficacy. Several related phase III trials are currently underway, and the results are highly anticipated.

Conclusions

The advent of the era of targeted and immune therapies, along with continuous innovations in technologies and methods, brings new hope to the treatment of HCC. However, more clinical and related basic research is still needed in the future to continually explore more effective, safer, and accessible modes of HCC conversion therapy and perioperative treatment, thereby improving the long-term survival of patients.

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Ethical Statement: The authors are accountable for all aspects of the work in ensuring that questions related to the accuracy or integrity of any part of the work are appropriately investigated and resolved.

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References

1. Expert Consensus Group on Postoperative Adjuvant Therapy for Liver Cancer, Chinese Medical Doctor Association of Surgeons, Chinese Anti-Cancer Association Liver Cancer Committee, Society of Oncology of Chinese Medical Association Liver Cancer Group. Expert consensus on postoperative adjuvant therapy for liver cancer in China (2023 Edition). Chin J Dig Surg 2023;22:437-48.
2. Bureau of Medical Administration and Medical Governance, National Health Commission of the People's Republic of China. Guidelines for the diagnosis and treatment of primary liver cancer (2022 Edition). Chin J Dig Surg 2022;21:143-68.
3. Xia YX, Zhang F, Li XC, et al. Surgical Treatment Analysis of 10,966 Cases of Primary Liver Cancer. Chin J Surg 2021;59:6-17.
4. Park JW, Chen M, Colombo M, et al. Global patterns of hepatocellular carcinoma management from diagnosis to death: the BRIDGE Study. Liver Int 2015;35:2155-66.
5. Cheng AL, Qin S, Ikeda M, et al. Updated efficacy and safety data from IMbrave150: Atezolizumab plus bevacizumab vs. sorafenib for unresectable hepatocellular carcinoma. J Hepatol 2022;76:862-73.
6. Yang SZ, Feng XB, Dong JH. Precision surgical concept guided liver cancer surgery. J Precis Med 2018;33:189-192,196.
7. Aloia TA. Associating Liver Partition and Portal Vein Ligation for Staged Hepatectomy: Portal Vein Embolization Should Remain the Gold Standard. JAMA Surg 2015;150:927-8.
8. Piron L, Deshayes E, Escal L, et al. Portal vein embolization: present and future. Bull Cancer 2017;104:407-16.
9. Dupré A, Hitier M, Peyrat P, et al. Associating portal embolization and artery ligation to induce rapid liver regeneration in staged hepatectomy. Br J Surg 2015;102:1541-50.
10. Glantzounis GK, Tokidis E, Basourakos SP, et al. The role of portal vein embolization in the surgical management of primary hepatobiliary cancers. A systematic review. Eur J Surg Oncol 2017;43:32-41.
11. Schnitzbauer AA, Lang SA, Goessmann H, et al. Right portal vein ligation combined with in situ splitting induces rapid left lateral liver lobe hypertrophy enabling 2-staged extended right hepatic resection in small-for-size settings. Ann Surg 2012;255:405-14.
12. Wang Z, Peng Y, Hu J, et al. Associating Liver Partition and Portal Vein Ligation for Staged Hepatectomy for Unresectable Hepatitis B Virus-related Hepatocellular Carcinoma: A Single Center Study of 45 Patients. Ann Surg 2020;271:534-41.
13. Peng Y, Wang Z, Qu X, et al. Transcatheter arterial embolization-salvaged ALPPS, a novel ALPPS procedure especially for patients with hepatocellular carcinoma and severe fibrosis/cirrhosis. Hepatobiliary Surg Nutr 2022;11:504-14.
14. Li PP, Huang G, Jia NY, et al. Associating liver partition and portal vein ligation for staged hepatectomy versus sequential transarterial chemoembolization and portal vein embolization in staged hepatectomy for HBV-related hepatocellular carcinoma: a randomized comparative study.

Hepatobiliary Surg Nutr 2022;11:38-51.
15. Ren Z, Xu J, Bai Y, et al. Sintilimab plus a bevacizumab biosimilar (IBI305) versus sorafenib in unresectable hepatocellular carcinoma (ORIENT-32): a randomised, open-label, phase 2-3 study. Lancet Oncol 2021;22:977-90.
16. Qin S, Chan SL, Gu S, et al. Camrelizumab plus rivoceranib versus sorafenib as first-line therapy for unresectable hepatocellular carcinoma (CARES-310): a randomised, open-label, international phase 3 study. Lancet 2023;402:1133-46.
17. Finn RS, Qin S, Ikeda M, et al. Atezolizumab plus Bevacizumab in Unresectable Hepatocellular Carcinoma. N Engl J Med 2020;382:1894-905.
18. Shiozaki H, Furukawa K, Haruki K, et al. A Multidisciplinary Treatment Strategy With Conversion Surgery for Hepatocellular Carcinoma. Anticancer Res 2023;43:1761-6.
19. Hidaka M, Hara T, Soyama A, et al. The Outcome of Conversion Liver Resection Surgery by Lenvatinib Treatment: A Single Center Experience. Anticancer Res 2022;42:3049-54.
20. Wang L, Wang H, Cui Y, et al. Sintilimab plus Lenvatinib conversion therapy for intermediate/locally advanced hepatocellular carcinoma: A phase 2 study. Front Oncol 2023;13:1115109.
21. Song T. A prospective, single-arm, phase II clinical study of tislelizumab in combination with lenvatinib for perioperative treatment of resectable primary hepatocellular carcinoma with high risk of recurrence. J Clin Oncol 2023;41:e16218.
22. Yi Y, Sun BY, Weng JL, et al. Lenvatinib plus anti-PD-1 therapy represents a feasible conversion resection strategy for patients with initially unresectable hepatocellular carcinoma: A retrospective study. Front Oncol 2022;12:1046584.
23. Zhang W, Tong S, Hu B, et al. Lenvatinib plus anti-PD-1 antibodies as conversion therapy for patients with unresectable intermediate-advanced hepatocellular carcinoma: a single-arm, phase II trial. J Immunother Cancer 2023;11:e007366.
24. Zhu XD, Huang C, Shen YH, et al. Hepatectomy After Conversion Therapy Using Tyrosine Kinase Inhibitors Plus Anti-PD-1 Antibody Therapy for Patients with Unresectable Hepatocellular Carcinoma. Ann Surg Oncol 2023;30:2782-90.
25. Xu B, Zhu XD, Shen YH, et al. Criteria for identifying potentially resectable patients with initially oncologically unresectable hepatocellular carcinoma before treatment with lenvatinib plus an anti-PD-1 antibody. Front Immunol 2022;13:1016736.
26. Wang L, Zhao Y, Zhang T, et al. 141P Efficacy and safety of tislelizumab combined with targeted therapy as first-line treatment for unresectable hepatocellular carcinoma: a real-world study. Immuno Oncol Tech 2022;16:100253.
27. Zhu XD, Huang C, Shen YH, et al. Downstaging and Resection of Initially Unresectable Hepatocellular Carcinoma with Tyrosine Kinase Inhibitor and Anti-PD-1 Antibody Combinations. Liver Cancer 2021;10:320-9.
28. Ho WJ, Sharma G, Zhu Q, et al. Integrated immunological analysis of a successful conversion of locally advanced hepatocellular carcinoma to resectability with neoadjuvant therapy. J Immunother Cancer 2020;8:e000932.
29. Zhang WW, Hu BY, Han J, et al. Preliminary report on the study of combination therapy using PD-1 inhibitors and multi-target tyrosine kinase inhibitors for advanced-stage hepatocellular carcinoma. Chin J Hepatobiliary Surg 2020;26:947-48.
30. Shimose S, Iwamoto H, Shirono T, et al. The impact of curative conversion therapy aimed at a cancer-free state in patients with hepatocellular carcinoma treated with atezolizumab plus bevacizumab. Cancer Med 2023;12:12325-35.
31. Miyata T, Sugi K, Horino T, et al. Conversion Surgery After Atezolizumab Plus Bevacizumab for Primary and Peritoneal Metastasis After Hepatocellular Carcinoma Rupture. Anticancer Res 2023;43:943-7.
32. Sun H, Zhu X, Gao Q, et al. 711P Sintilimab combined with bevacizumab biosimilar as a conversion therapy in potentially resectable intermediate stage hepatocellular carcinoma (HCC): a phase II trial. Ann Oncol 2022;33:S867-S868.
33. Fukunaga A, Takata K, Itoh S, et al. Complete tumor necrosis confirmed by conversion hepatectomy after atezolizumab-bevacizumab treatment for advanced-stage hepatocellular carcinoma with lung metastasis. Clin J Gastroenterol 2023;16:224-8.
34. Kudo M. A Novel Treatment Strategy for Patients with Intermediate-Stage HCC Who Are Not Suitable for TACE: Upfront Systemic Therapy Followed by Curative Conversion. Liver Cancer 2021;10:539-44.
35. Chinese Society of Interventional Radiology, Clinical Diagnosis and Treatment Guidelines Committee of Chinese Medical Doctor Association. Clinical practice guidelines for transarterial chemoembolization (TACE)

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treatment of hepatocellular carcinoma (2023 Edition). Chin Med J 2023;103:2674-94.
36. Chinese Anti-Cancer Association Committee of Tumor Intervention. Clinical consensus on the use of drug-eluting beads for the treatment of unresectable primary liver cancer. Chin J Radiol 2022;56:349-55.
37. Zhang ZF, Luo YJ, Lu Q, et al. Conversion therapy and suitable timing for subsequent salvage surgery for initially unresectable hepatocellular carcinoma: What is new? World J Clin Cases 2018;6:259-73.
38. Zhang Y, Huang G, Wang Y, et al. Is Salvage Liver Resection Necessary for Initially Unresectable Hepatocellular Carcinoma Patients Downstaged by Transarterial Chemoembolization? Ten Years of Experience. Oncologist 2016;21:1442-9.
39. Allard MA, Sebagh M, Ruiz A, et al. Does pathological response after transarterial chemoembolization for hepatocellular carcinoma in cirrhotic patients with cirrhosis predict outcome after liver resection or transplantation? J Hepatol 2015;63:83-92.
40. Chen S, Peng Z, Zhang Y, et al. Lack of Response to Transarterial Chemoembolization for Intermediate-Stage Hepatocellular Carcinoma: Abandon or Repeat? Radiology 2021;298:680-92.
41. Lyu N, Wang X, Li JB, et al. Arterial Chemotherapy of Oxaliplatin Plus Fluorouracil Versus Sorafenib in Advanced Hepatocellular Carcinoma: A Biomolecular Exploratory, Randomized, Phase III Trial (FOHAIC-1). J Clin Oncol 2022;40:468-80.
42. Li B, Qiu J, Zheng Y, et al. Conversion to Resectability Using Transarterial Chemoembolization Combined With Hepatic Arterial Infusion Chemotherapy for Initially Unresectable Hepatocellular Carcinoma. Ann Surg Open 2021;2:e057.
43. He C, Ge N, Wang X, et al. Conversion Therapy of Large Unresectable Hepatocellular Carcinoma With Ipsilateral Portal Vein Tumor Thrombus Using Portal Vein Embolization Plus Transcatheter Arterial Chemoembolization. Front Oncol 2022;12:923566.
44. Zhu HD, Li HL, Huang MS, et al. Transarterial chemoembolization with PD-(L)1 inhibitors plus molecular targeted therapies for hepatocellular carcinoma (CHANCE001). Signal Transduct Target Ther 2023;8:58.
45. Peng Z, Fan W, Zhu B, et al. Lenvatinib Combined With Transarterial Chemoembolization as First-Line Treatment for Advanced Hepatocellular Carcinoma: A Phase III, Randomized Clinical Trial (LAUNCH). J Clin Oncol 2023;41:117-27.
46. Wu JY, Wu JY, Li YN, et al. Lenvatinib combined with anti-PD-1 antibodies plus transcatheter arterial chemoembolization for neoadjuvant treatment of resectable hepatocellular carcinoma with high risk of recurrence: A multicenter retrospective study. Front Oncol 2022;12:985380.
47. Cao Y, Zhang X, Yue Y, et al. Donafenib combined with camrelizumab and transarterial chemoembolization (TACE) as conversion therapy for unresectable hepatocellular carcinoma (uHCC): a prospective, single-arm, phase II study. J Clin Oncol 2023;41:e16203.
48. Li QJ, He MK, Chen HW, et al. Hepatic Arterial Infusion of Oxaliplatin, Fluorouracil, and Leucovorin Versus Transarterial Chemoembolization for Large Hepatocellular Carcinoma: A Randomized Phase III Trial. J Clin Oncol 2022;40:150-60.
49. He M, Li Q, Zou R, et al. Sorafenib Plus Hepatic Arterial Infusion of Oxaliplatin, Fluorouracil, and Leucovorin vs Sorafenib Alone for Hepatocellular Carcinoma With Portal Vein Invasion: A Randomized Clinical Trial. JAMA Oncol 2019;5:953-60.
50. Zheng K, Zhu X, Fu S, et al. Sorafenib Plus Hepatic Arterial Infusion Chemotherapy versus Sorafenib for Hepatocellular Carcinoma with Major Portal Vein Tumor Thrombosis: A Randomized Trial. Radiology 2022;303:455-64.
51. He MK, Liang RB, Zhao Y, et al. Lenvatinib, toripalimab, plus hepatic arterial infusion chemotherapy versus lenvatinib alone for advanced hepatocellular carcinoma. Ther Adv Med Oncol 2021;13:17588359211002720.
52. Zhang TQ, Geng ZJ, Zuo MX, et al. Camrelizumab (a PD-1 inhibitor) plus apatinib (an VEGFR-2 inhibitor) and hepatic artery infusion chemotherapy for hepatocellular carcinoma in Barcelona Clinic Liver Cancer stage C (TRIPLET): a phase II study. Signal Transduct Target Ther 2023;8:413.
53. Liu D, Mu H, Liu C, et al. Hepatic artery infusion chemotherapy (HAIC) combined with sintilimab and bevacizumab biosimilar (IBI305) for initial unresectable hepatocellular carcinoma (HCC): a prospective, single-arm phase II trial. J Clin Oncol 2022;40:abstract 4073.
54. Dong W, Zhang SY, Huo ZN, et al. Lenvatinib in combination with PD-1 inhibitor and hepatic arterial infusion chemotherapy (HAIC) for patients with potentially resectable hepatocellular carcinoma: a retrospective analysis. J Clin Oncol 2023;41:e16160.
55. Tan K, He X, Zhang H, et al. Efficacy and safety of tislelizumab combined with lenvatinib and FOLFOX4-

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HAIC in conversion therapy of middle-advanced stage hepatocellular carcinoma (HCC): a real-world retrospective study. J Clin Oncol 2023;41:e16137.
56. Zhang W, Gao W, Liu C, et al. 999P Donafenib combined with hepatic artery infusion chemotherapy (HAIC) and sintilimab for unresectable hepatocellular carcinoma (uHCC): a prospective, single-arm phase II trial (DoHAICs study). Ann Oncol 2023;34:S613.
57. Lee IJ, Kim JW, Han KH, et al. Concurrent chemoradiotherapy shows long-term survival after conversion from locally advanced to resectable hepatocellular carcinoma. Yonsei Med J 2014;55:1489-97.
58. Lee HS, Choi GH, Choi JS, et al. Surgical resection after down-staging of locally advanced hepatocellular carcinoma by localized concurrent chemoradiotherapy. Ann Surg Oncol 2014;21:3646-53.
59. Byun HK, Kim HJ, Im YR, et al. Dose escalation by intensity modulated radiotherapy in liver-directed concurrent chemoradiotherapy for locally advanced BCLC stage C hepatocellular carcinoma. Radiother Oncol 2019;133:1-8.
60. Chiang CL, Chiu KWH, Chan KSK, et al. Sequential transarterial chemoembolisation and stereotactic body radiotherapy followed by immunotherapy as conversion therapy for patients with locally advanced, unresectable hepatocellular carcinoma (START-FIT): a single-arm, phase 2 trial. Lancet Gastroenterol Hepatol 2023;8:169-78.
61. Wang Q, Ji X, Sun J, et al. Comparison of stereotactic body radiotherapy with and without lenvatinib for advanced hepatocellular carcinoma: a propensity score analysis. J Cancer Res Clin Oncol 2023;149:7441-52.
62. Fan J, Tang ZY, Yu YQ, et al. Improved survival with resection after transcatheter arterial chemoembolization (TACE) for unresectable hepatocellular carcinoma. Dig Surg 1998;15:674-8.
63. Kulik LM, Atassi B, van Holsbeeck L, et al. Yttrium-90 microspheres (TheraSphere) treatment of unresectable hepatocellular carcinoma: downstaging to resection, RFA and bridge to transplantation. J Surg Oncol 2006;94:572-86.
64. Lewandowski RJ, Kulik LM, Riaz A, et al. A comparative analysis of transarterial downstaging for hepatocellular carcinoma: chemoembolization versus radioembolization. Am J Transplant 2009;9:1920-8.
65. Shindoh J, Kawamura Y, Kobayashi Y, et al. Prognostic Impact of Surgical Intervention After Lenvatinib Treatment for Advanced Hepatocellular Carcinoma. Ann Surg Oncol 2021;28:7663-72.
66. Kaneko S, Tsuchiya K, Yasui Y, et al. Conversion surgery for hepatocellular carcinoma after tyrosine kinase inhibitor treatment. JGH Open 2022;6:301-8.
67. Yarchoan M, Zhu Q, Durham JN, et al. Feasibility and efficacy of neoadjuvant cabozantinib and nivolumab in patients with borderline resectable or locally advanced hepatocellular carcinoma (HCC). J Clin Oncol 2021;39:abstract 335.
68. Kaseb AO, Tran Cao HS, Mohamed YI, et al. Final results of a randomized, open label, perioperative phase II study evaluating nivolumab alone or nivolumab plus ipilimumab in patients with resectable HCC. J Clin Oncol 2020;38 abstract 4599.
69. Finn RS, Kudo M, Merle P, et al. LBA34 Primary results from the phase III LEAP-002 study: lenvatinib plus pembrolizumab versus lenvatinib as first-line (1L) therapy for advanced hepatocellular carcinoma (aHCC). Ann Oncol 2022;33:S1401.
70. Dhir M, Sasson AR. Surgical Management of Liver Metastases From Colorectal Cancer. J Oncol Pract 2016;12:33-9.
71. Cao Y, Tang H, Hu B, et al. Comparison of survival benefit between salvage surgery after conversion therapy versus surgery alone for hepatocellular carcinoma with portal vein tumor thrombosis: a propensity score analysis. HPB (Oxford) 2023;25:775-87.
72. Bertacco A, Vitale A, Mescoli C, et al. Sorafenib treatment has the potential to downstage advanced hepatocellular carcinoma before liver resection. Per Med 2020;17:83-7.
73. Kim TS, Kim JH, Kim BH, et al. Complete response of advanced hepatocellular carcinoma to sorafenib: another case and a comprehensive review. Clin Mol Hepatol 2017;23:340-6.
74. Irtan S, Chopin-Laly X, Ronot M, et al. Complete regression of locally advanced hepatocellular carcinoma induced by sorafenib allowing curative resection. Liver Int 2011;31:740-3.
75. Cleary JM, Tanabe KT, Lauwers GY, et al. Hepatic toxicities associated with the use of preoperative systemic therapy in patients with metastatic colorectal adenocarcinoma to the liver. Oncologist 2009;14:1095-105.
76. Kaseb AO, Vence L, Blando J, et al. Immunologic Correlates of Pathologic Complete Response to Preoperative Immunotherapy in Hepatocellular Carcinoma. Cancer Immunol Res 2019;7:1390-5.
77. Kaseb A, Vence L, Blando J, et al. Randomized, open-

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label, perioperative phase II study evaluating nivolumab alone versus nivolumab plus ipilimumab in patients with resectable HCC. J Clin Oncol 2019;37:abstract 185.
78. Shen YH, Huang C, Zhu XD, et al. The Safety Profile of Hepatectomy Following Preoperative Systemic Therapy with Lenvatinib Plus Anti-PD-1 Antibodies Versus Hepatectomy Alone in Patients With Hepatocellular Carcinoma. Ann Surg Open 2022;3:e163.
79. Luo L, He Y, Zhu G, et al. Hepatectomy After Conversion Therapy for Initially Unresectable HCC: What is the Difference? J Hepatocell Carcinoma 2022;9:1353-68.
80. Paye F, Jagot P, Vilgrain V, et al. Preoperative chemoembolization of hepatocellular carcinoma: a comparative study. Arch Surg 1998;133:767-72.
81. Li C, Wang MD, Lu L, et al. Preoperative transcatheter arterial chemoembolization for surgical resection of huge hepatocellular carcinoma (≥10 cm): a multicenter propensity matching analysis. Hepatol Int 2019;13:736-47.
82. Majno PE, Adam R, Bismuth H, et al. Influence of preoperative transarterial lipiodol chemoembolization on resection and transplantation for hepatocellular carcinoma in patients with cirrhosis. Ann Surg 1997;226:688-701; discussion 701-3.
83. Tustumi F, Ernani L, Coelho FF, et al. Preoperative strategies to improve resectability for hepatocellular carcinoma: a systematic review and meta-analysis. HPB (Oxford) 2018;20:1109-18.
84. Li N, Feng S, Xue J, et al. Hepatocellular carcinoma with main portal vein tumor thrombus: a comparative study comparing hepatectomy with or without neoadjuvant radiotherapy. HPB (Oxford) 2016;18:549-56.
85. Zhou H, Song T. Conversion therapy and maintenance therapy for primary hepatocellular carcinoma. Biosci Trends 2021;15:155-60.
86. Ducreux M, Abou-Alfa GK, Bekaii-Saab T, et al. The management of hepatocellular carcinoma. Current expert opinion and recommendations derived from the 24th ESMO/World Congress on Gastrointestinal Cancer, Barcelona, 2022. ESMO Open 2023;8:101567.
87. Yoo C, Kim JH, Ryu MH, et al. Clinical Outcomes with Multikinase Inhibitors after Progression on First-Line Atezolizumab plus Bevacizumab in Patients with Advanced Hepatocellular Carcinoma: A Multinational Multicenter Retrospective Study. Liver Cancer 2021;10:107-14.
88. Wong JSL, Kwok GGW, Tang V, et al. Ipilimumab and nivolumab/pembrolizumab in advanced hepatocellular carcinoma refractory to prior immune checkpoint inhibitors. J Immunother Cancer 2021;9:e001945.
89. Bruix J, Qin S, Merle P, et al. Regorafenib for patients with hepatocellular carcinoma who progressed on sorafenib treatment (RESORCE): a randomised, double-blind, placebo-controlled, phase 3 trial. Lancet 2017;389:56-66.
90. Lim HY, Kim YJ, Huang YH, et al. 1009P Regorafenib in patients (pts) with unresectable hepatocellular carcinoma (uHCC) in real-world practice in Asia: interim results from the observational REFINE study. Ann Oncol 2020;31:S699.
91. Qin S, Li Q, Gu S, et al. Apatinib as second-line or later therapy in patients with advanced hepatocellular carcinoma (AHELP): a multicentre, double-blind, randomised, placebo-controlled, phase 3 trial. Lancet Gastroenterol Hepatol 2021;6:559-68.
92. Qin S, Ren Z, Meng Z, et al. Camrelizumab in patients with previously treated advanced hepatocellular carcinoma: a multicentre, open-label, parallel-group, randomised, phase 2 trial. Lancet Oncol 2020;21:571-80.
93. Edeline J, Merle P, Fang W, et al. Clinical outcomes associated with tislelizumab in patients with advanced hepatocellular carcinoma who have been previously treated with sorafenib or lenvatinib in RATIONALE-208. J Clin Oncol 2022;40:abstract 420.
94. Qin S, Chen Z, Fang W, et al. Pembrolizumab Versus Placebo as Second-Line Therapy in Patients From Asia With Advanced Hepatocellular Carcinoma: A Randomized, Double-Blind, Phase III Trial. J Clin Oncol 2023;41:1434-43.
95. Li S, Zhong C, Li Q, et al. Neoadjuvant transarterial infusion chemotherapy with FOLFOX could improve outcomes of resectable BCLC stage A/B hepatocellular carcinoma patients beyond Milan criteria: an interim analysis of a multicenter, phase 3, randomized, controlled clinical trial. J Clin Oncol 2021;39:abstract 4008.
96. Wei X, Jiang Y, Zhang X, et al. Neoadjuvant Three-Dimensional Conformal Radiotherapy for Resectable Hepatocellular Carcinoma With Portal Vein Tumor Thrombus: A Randomized, Open-Label, Multicenter Controlled Study. J Clin Oncol 2019;37:2141-51.
97. Kamiyama T, Nakanishi K, Yokoo H, et al. Efficacy of preoperative radiotherapy to portal vein tumor thrombus in the main trunk or first branch in patients with hepatocellular carcinoma. Int J Clin Oncol 2007;12:363-8.
98. Wu F, Chen B, Dong D, et al. Phase 2 Evaluation of Neoadjuvant Intensity-Modulated Radiotherapy in Centrally Located Hepatocellular Carcinoma:

© AME Publishing Company.

A Nonrandomized Controlled Trial. JAMA Surg 2022;157:1089-96.
99. Zhou J, Fan J, Gu FM, et al. A phase II/III study of camrelizumab plus apatinib as perioperative treatment of resectable hepatocellular carcinoma at intermediate-high risk of recurrence: primary results of major pathologic response from phase II stage. J Clin Oncol 2023;41:abstract 4126.
100. Chan AWH, Zhong J, Berhane S, et al. Development of pre and post-operative models to predict early recurrence of hepatocellular carcinoma after surgical resection. J Hepatol 2018;69:1284-93.
101. Wu JC, Huang YH, Chau GY, et al. Risk factors for early and late recurrence in hepatitis B-related hepatocellular carcinoma. J Hepatol 2009;51:890-7.
102. Imamura H, Matsuyama Y, Tanaka E, et al. Risk factors contributing to early and late phase intrahepatic recurrence of hepatocellular carcinoma after hepatectomy. J Hepatol 2003;38:200-7.
103. Wang Z, Ren Z, Chen Y, et al. Adjuvant Transarterial Chemoembolization for HBV-Related Hepatocellular Carcinoma After Resection: A Randomized Controlled Study. Clin Cancer Res 2018;24:2074-81.
104. Wei W, Jian PE, Li SH, et al. Adjuvant transcatheter arterial chemoembolization after curative resection for hepatocellular carcinoma patients with solitary tumor and microvascular invasion: a randomized clinical trial of efficacy and safety. Cancer Commun (Lond) 2018;38:61.
105. Li SH, Mei J, Cheng Y, et al. Postoperative Adjuvant Hepatic Arterial Infusion Chemotherapy With FOLFOX in Hepatocellular Carcinoma With Microvascular Invasion: A Multicenter, Phase III, Randomized Study. J Clin Oncol 2023;41:1898-908.
106. Chen Q, Shu C, Laurence AD, et al. Effect of Huaier granule on recurrence after curative resection of HCC: a multicentre, randomised clinical trial. Gut 2018;67:2006-16.
107. Zhou J, Sun HC, Huang ZY, et al. Adjuvant lenvatinib after radical resection in patients with hepatocellular carcinoma: preliminary analysis of a prospective, multi-center, single-arm study. J Clin Oncol 2022;40:e16158.
108. Chen B, Wu JX, Cheng SH, et al. Phase 2 Study of Adjuvant Radiotherapy Following Narrow-Margin Hepatectomy in Patients With HCC. Hepatology 2021;74:2595-604.
109. Lee JH, Lee JH, Lim YS, et al. Adjuvant immunotherapy with autologous cytokine-induced killer cells for hepatocellular carcinoma. Gastroenterology 2015;148:1383-91.e6.
110. Huang G, Li PP, Lau WY, et al. Antiviral Therapy Reduces Hepatocellular Carcinoma Recurrence in Patients With Low HBV-DNA Levels: A Randomized Controlled Trial. Ann Surg 2018;268:943-54.
111. Wu J, Yin Z, Cao L, et al. Adjuvant pegylated interferon therapy improves the survival outcomes in patients with hepatitis-related hepatocellular carcinoma after curative treatment: A meta-analysis. Medicine (Baltimore) 2018;97:e11295.
112. Qin S, Chen M, Cheng AL, et al. Atezolizumab plus bevacizumab versus active surveillance in patients with resected or ablated high-risk hepatocellular carcinoma (IMbrave050): a randomised, open-label, multicentre, phase 3 trial. Lancet 2023;402:1835-47.
113. Yopp A, Kudo M, Chen M, et al. LBA39 Updated efficacy and safety data from IMbrave050: Phase III study of adjuvant atezolizumab (atezo) + bevacizumab (bev) vs active surveillance in patients (pts) with resected or ablated high-risk hepatocellular carcinoma (HCC). Ann Oncol 2024;35:S1230.
114. Travis WD, Dacic S, Wistuba I, et al. IASLC Multidisciplinary Recommendations for Pathologic Assessment of Lung Cancer Resection Specimens After Neoadjuvant Therapy. J Thorac Oncol 2020;15:709-40.
115. Stein JE, Lipson EJ, Cottrell TR, et al. Pan-Tumor Pathologic Scoring of Response to PD-(L)1 Blockade. Clin Cancer Res 2020;26:545-51.
116. Huang C, Zhu XD, Shen YH, et al. Radiographic and α-fetoprotein response predict pathologic complete response to immunotherapy plus a TKI in hepatocellular carcinoma: a multicenter study. BMC Cancer 2023;23:416.
117. Lee YJ, Lee JM, Lee JS, et al. Hepatocellular carcinoma: diagnostic performance of multidetector CT and MR imaging-a systematic review and meta-analysis. Radiology 2015;275:97-109.
118. Seymour L, Bogaerts J, Perrone A, et al. iRECIST: guidelines for response criteria for use in trials testing immunotherapeutics. Lancet Oncol 2017;18:e143-52.
119. Xu B, Dong SY, Bai XL, et al. Tumor Radiomic Features on Pretreatment MRI to Predict Response to Lenvatinib plus an Anti-PD-1 Antibody in Advanced Hepatocellular Carcinoma: A Multicenter Study. Liver Cancer 2023;12:262-76.
120. Kudo M, Finn RS, Qin S, et al. Lenvatinib versus sorafenib in first-line treatment of patients with unresectable hepatocellular carcinoma: a randomised phase 3 non-

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inferiority trial. Lancet 2018;391:1163-73.
121. Qin S, Bi F, Gu S, et al. Donafenib Versus Sorafenib in First-Line Treatment of Unresectable or Metastatic Hepatocellular Carcinoma: A Randomized, Open-Label, Parallel-Controlled Phase II-III Trial. J Clin Oncol 2021;39:3002-11.
122. Wang DY, Salem JE, Cohen JV, et al. Fatal Toxic Effects Associated With Immune Checkpoint Inhibitors: A Systematic Review and Meta-analysis. JAMA Oncol 2018;4:1721-8.
123. Xu J, Shen J, Gu S, et al. Camrelizumab in Combination with Apatinib in Patients with Advanced Hepatocellular Carcinoma (RESCUE): A Nonrandomized, Open-label, Phase II Trial. Clin Cancer Res 2021;27:1003-11.
124. Kelley RK, Rimassa L, Cheng AL, et al. Cabozantinib plus atezolizumab versus sorafenib for advanced hepatocellular carcinoma (COSMIC-312): a multicentre, open-label, randomised, phase 3 trial. Lancet Oncol 2022;23:995-1008.
125. Finn RS, Ikeda M, Zhu AX, et al. Phase Ib Study of Lenvatinib Plus Pembrolizumab in Patients With Unresectable Hepatocellular Carcinoma. J Clin Oncol 2020;38:2960-70.
126. U.S. Department of Health and Human Services. Common Terminology Criteria for Adverse Events (CTCAE) Version 5.0. (2017-11-27) [2023-11-01]. CTCAE Version 5.0 Quick Reference.
127. Chinese Society of Clinical Oncology (CSCO) Guideline Committee. CSCO guidelines for the management of toxicity associated with immune checkpoint inhibitors 2021. Beijing: People's Medical Publishing House; 2021:1-140.
128. Chinese Medical Association's Professional Committee on Hepatocellular Carcinoma. Expert consensus on the clinical application of molecular targeted drugs for hepatocellular carcinoma in China (2022 Edition). Chin Med J 2022;102:2655-68.
129. Guo DZ, Zhang SY, Dong SY, et al. Prognostic model for predicting outcome and guiding treatment decision for unresectable hepatocellular carcinoma treated with lenvatinib monotherapy or lenvatinib plus immunotherapy. Front Immunol 2023;14:1141199.
130. Guo DZ, Zhang SY, Dong SY, et al. Circulating immune index predicting the prognosis of patients with hepatocellular carcinoma treated with lenvatinib and immunotherapy. Front Oncol 2023;13:1109742.
131. Huang C, Xu B, Zhu XD, et al. Peripheral naïve CD8(+) T cells as a predictive biomarker of response to lenvatinib plus an anti-PD-1 antibody in advanced hepatocellular carcinoma: a biomarker study. Cancer Commun (Lond) 2022;42:1226-30.
132. Bo Z, Chen B, Zhao Z, et al. Prediction of Response to Lenvatinib Monotherapy for Unresectable Hepatocellular Carcinoma by Machine Learning Radiomics: A Multicenter Cohort Study. Clin Cancer Res 2023;29:1730-40.
133. Tada T, Kumada T, Hiraoka A, et al. Neutrophil-lymphocyte ratio predicts early outcomes in patients with unresectable hepatocellular carcinoma treated with atezolizumab plus bevacizumab: a multicenter analysis. Eur J Gastroenterol Hepatol 2022;34:698-706.
134. Eso Y, Takeda H, Taura K, et al. Pretreatment Neutrophil-to-Lymphocyte Ratio as a Predictive Marker of Response to Atezolizumab Plus Bevacizumab for Hepatocellular Carcinoma. Curr Oncol 2021;28:4157-66.
135. Ochi H, Kurosaki M, Joko K, et al. Usefulness of neutrophil-to-lymphocyte ratio in predicting progression and survival outcomes after atezolizumab-bevacizumab treatment for hepatocellular carcinoma. Hepatol Res 2023;53:61-71.
136. Matoya S, Suzuki T, Matsuura K, et al. The neutrophil-to-lymphocyte ratio at the start of the second course during atezolizumab plus bevacizumab therapy predicts therapeutic efficacy in patients with advanced hepatocellular carcinoma: A multicenter analysis. Hepatol Res 2023;53:511-21.
137. Wu YL, Fulgenzi CAM, D'Alessio A, et al. Neutrophil-to-Lymphocyte and Platelet-to-Lymphocyte Ratios as Prognostic Biomarkers in Unresectable Hepatocellular Carcinoma Treated with Atezolizumab plus Bevacizumab. Cancers (Basel) 2022;14:5834.
138. Wang JH, Chen YY, Kee KM, et al. The Prognostic Value of Neutrophil-to-Lymphocyte Ratio and Platelet-to-Lymphocyte Ratio in Patients with Hepatocellular Carcinoma Receiving Atezolizumab Plus Bevacizumab. Cancers (Basel) 2022;14:343.
139. Zhu AX, Abbas AR, de Galarreta MR, et al. Molecular correlates of clinical response and resistance to atezolizumab in combination with bevacizumab in advanced hepatocellular carcinoma. Nat Med 2022;28:1599-611.
140. Zhang W, Gong C, Peng X, et al. Serum Concentration of CD137 and Tumor Infiltration by M1 Macrophages Predict the Response to Sintilimab plus Bevacizumab Biosimilar in Advanced Hepatocellular Carcinoma Patients. Clin Cancer Res 2022;28:3499-508.

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141. Yamauchi R, Ito T, Yoshio S, et al. Serum osteopontin predicts the response to atezolizumab plus bevacizumab in patients with hepatocellular carcinoma. J Gastroenterol 2023;58:565-74.

142. Wang Q, Xia D, Bai W, et al. Development of a prognostic score for recommended TACE candidates with hepatocellular carcinoma: A multicentre observational study. J Hepatol 2019;70:893-903.

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